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A WEEKLY JOURNAL:

CONDUCTED BY CHARLES DICKENS.

N^o. 19.]

SATURDAY, AUGUST 3, 1850.

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THE LAST OF A LONG LINE.

IN TWO CHAPTERS.—CHAPTER I.

Sir Roger Rockville of Rockville was the last of a very long line. It extended from the Norman Conquest to the present century. His first known ancestor came over with William, and must have been a man of some mark, either of bone and sinew, or of brain, for he obtained what the Americans would call a prime location. As his name does not occur in the Roll of Battle Abbey, he was, of course, not of a very high Norman extraction; but he had done enough, it seems, in the way of knocking down Saxons, to place himself on a considerable eminence in this kingdom. The centre of his domains was conspicuous far over the country, through a high range of rock overhanging one of the sweetest rivers in England. On one hand lay a vast tract of rich marsh land, capable, as society advanced, of being converted into meadows; and on the other, as extensive moorlands, finely undulating, and abounding with woods and deer.

Here the original Sir Roger built his castle on the summit of the range of rock, with huts for his followers; and became known directly all over the country of Sir Roger de Rockville, or Sir Roger of the hamlet on the Rock. Sir Roger, no doubt, was a mighty hunter before the lord of the feudal district: it is certain that his descendants were. For generations they led a jolly life at Rockville, and were always ready to exchange the excitement of the chase for a bit of civil war. Without that the country would have grown dull, and ale and venison lost their flavour. There was no gay London in those days, and a good brisk skirmish with their neighbours in helm and hauberk was the way of spending their season. It was their parliamentary debate, and was necessary to thin their woods. Protection and Free Trade were as much the great topics of interest as they are now, only they did not

trouble themselves so much about Corn bills. Their bills were of good steel, and their protective measures were arrows a cloth-yard long. Protection meant a good suit of mail; and a castle with its duly prescribed moats, bastions, portcullises, and donjon keep. Free Trade was a lively inroad into the neighbouring baron's lands, and the importation thence of goodly herds and flocks. Foreign cattle for home consumption was as *sticking an article* in their markets as in ours, only the blows were expended on one another's heads, instead of the heads of foreign bullocks—that is, bullocks from over the Welch or Scotch Marches, as from beyond the next brook.

Thus lived the Rockvilles for ages. In all the iron combats of those iron times they took care to have their quota. Whether it were Stephen against Matilda, or Richard against his father, or John against the barons; whether it were York or Lancaster, or Tudor or Stuart. The Rockvilles were to be found in the *mêlée*, and winning power and lands. So long as it required only stalwart frames and stout blows, no family cut a more conspicuous figure. The Rockvilles were at Bosworth Field. The Rockvilles fought in Ireland under Elizabeth. The Rockvilles were staunch defenders of the cause in the war of Charles I. with his Parliament. The Rockvilles even fought for James II. at the Boyne, when three-fourths of the most loyal of the English nobility and gentry had deserted him in disgust and indignation. But from that hour they had been less conspicuous.

The opposition to the successful party, that of William of Orange, of course brought them into disgrace: and though they were never molested on that account, they retired to their estate, and found it convenient to be as unobtrusive as possible. Thenceforward you heard no more of the Rockvilles in the national annals. They became only of consequence in their own district. They acted as magistrates. They served as high sheriffs. They were a substantial county family, and nothing more. Education and civilisation advanced; a wider and very different field of action and ambition opened upon the aristocracy of England. Our fleets and armies abroad, our legislature at home, law and the church, presented brilliant paths to the ambition of those thirsting for distinction, and the good things that follow it. But somehow the Rockvilles did not expand with this expansion. So long as it required only a figure of six feet high, broad shoulders, and a strong arm, they were a great and conspicuous race. But when the head became the member most in request, they ceased to go ahead. Younger sons, it is true, served in army and in navy, and filled the

family pulpit, but they produced no generals, no admirals, no archbishops. The Rockvilles of Rockville were very conservative, very exclusive, and very stereotype. Other families grew poor, and enriched themselves again by marrying plebeian heiresses. New families grew up out of plebeian blood into greatness, and intermingled the vigour of their fresh earth with the attenuated aristocratic soil. Men of family became great lawyers, great statesmen, great prelates, and even great poets and philosophers. The Rockvilles remained high, proud, bigotted, and *borné*.

The Rockvilles married Rockvilles, or their first cousins, the Cesgvilles, simply to prevent property going out of the family. They kept the property together. They did not lose an acre, and they were a fine, tall, solemn race—and nothing more. What ailed them?

If you saw Sir Roger Rockville,—for there was an eternal Sir Roger filling his office of high sheriff,—he had a very fine carriage, and a very fine retinue in the most approved and splendid of antique costumes;—if you saw him sitting on the bench at quarter sessions, he was a tall, stately, and solemn man. If you saw Lady Rockville shopping, in her handsome carriage, with very handsomely attired servants; saw her at the county ball, or on the race-stand, she was a tall, aristocratic, and stately lady. That was in the last generation—the present could boast of no Lady Rockville.

Great outward respect was shown to the Rockvilles on account of the length of their descent, and the breadth of their acres. They were always, when any stranger asked about them, declared, with a serious and important air, to be a very ancient, honourable, and substantial family. “Oh! a great family are the Rockvilles, a very great family.”

But if you came to close quarters with the members of this great and highly distinguished family, you soon found yourself fundamentally astonished: you had a sensation come over you, as if you were trying, like Moses, to draw water from a rock, without his delegated power. There was a goodly outside of things before you, but nothing came of it. You talked, hoping to get talking in return, but you got little more than “noes” and “yeses,” and “oh! indeeds!” and “reallys,” and sometimes not even that, but a certain look of aristocratic dignity or dignification, that was meant to serve for all answers. There was a sort of resting on aristocratic oars or “sculls,” that were not to be too vulgarly handled. There was a feeling impressed on you, that eight hundred years of descent and ten thousand a-

year in landed income did not trouble themselves with the trifling things that gave distinction to lesser people—such as literature, fine arts, politics, and general knowledge. These were very well for those who had nothing else to pride themselves on, but for the Rockvilles—oh! certainly they were by no means requisite.

In fact, you found yourself, with a little variation, in the predicament of Cowper's people,

—— who spent their lives
In dropping buckets into empty wells,
And *growing tired* of drawing nothing up.

Who hasn't often come across these "dry wells" of society; solemn gulphs out of which you can pump nothing up? You know them; they are at your elbow every day in large and brilliant companies, and defy the best sucking-buckets ever invented to extract anything from them. But the Rockvilles were each and all of this adust description. It was a family feature, and they seemed, if either, rather proud of it. They must be so; for proud they were, amazingly proud; and they had nothing besides to be proud of, except their acres, and their ancestors.

But the fact was, they could not help it. It was become organic. They had acted the justice of peace, maintained the constitution against upstarts and manufacturers, signed warrants, supported the church and the house of correction, committed poachers, and then rested on the dignity of their ancestors for so many generations, that their skulls, brains, constitutions, and nervous systems, were all so completely moulded into that shape and baked into that mould, that a Rockville would be a Rockville to the end of time, if God and Nature would have allowed it. But such things wear out. The American Indians and the Australian nations wear out; they are not progressive, and as Nature abhors a vacuum, she does not forget the vacuum wherever it may be, whether in a hot desert, or in a cold and stately Rockville;—a very ancient, honourable, and substantial family that lies fallow till the thinking faculty literally dies out.

For several generations there had been symptoms of decay about the Rockville family. Not in its property, that was as large as ever; not in their personal stature and physical aspect. The Rockvilles continued, as they always had been, a tall and not bad-looking family. But they grew gradually less prolific. For a hundred and fifty years past there had seldom been more

than two, or at most three, children. There had generally been an heir to the estate, and another to the family pulpit, and sometimes a daughter married to some neighbouring squire. But Sir Roger's father had been an only child, and Sir Roger himself was an only child. The danger of extinction to the family, apparent as it was, had never induced Sir Roger to marry. At the time that we are turning our attention upon him, he had reached the mature age of sixty. Nobody believed that Sir Roger now would marry; he was the last, and likely to be, of his line.

It is worth while here to take a glance at Sir Roger and his estate. They wore a strange contrast. The one bore all the signs of progress, the other of a stereotyped feudality. The estate, which in the days of the first Sir Roger de Rockville had been half morass and half wilderness, was now cultivated to the pitch of British agricultural science. The marshlands beyond the river were one splendid expanse of richest meadows, yielding a rental of four solid pounds per acre. Over hill and dale on this side for miles, where formerly ran wild deer, and grew wild woodlands or furze-bushes, now lay excellent farms and hamlets, and along the ridge of the ancient cliffs rose the most magnificent woods. Woods, too, clothed the steep hill-sides, and swept down to the noble river, their very boughs hanging far out over its clear and rapid waters. In the midst of these fine woods stood Rockville Hall, the family seat of the Rockvilles. It reared its old brick walls above the towering mass of elms, and travellers at a distance recognised it for what it was, the mansion of an ancient and wealthy family.

The progress of England in arts, science, commerce, and manufacture, had carried Sir Roger's estate along with it. It was full of active and moneyed farmers, and flourished under modern influences. How lucky it would have been for the Rockville family had it done the same!

But amid this estate there was Sir Roger solitary, and the last of the line. He had grown well enough—there was nothing stunted about him, so far as you could see on the surface. In stature, he exceeded six feet. His colossal elms could not boast of a properer relative growth. He was as large a landlord, and as tall a justice of the peace, as you could desire; but, unfortunately, he was, after all, only the shell of a man. Like many of his veteran elms, there was a very fine stem, only it was hollow. There was a man, just with the rather awkward deficiency of a soul.

And it were no difficult task to explain, either, how this had come about. The Rockvilles saw plainly enough the necessity of manuring their lands, but they scorned the very idea of manuring their family. What! that most ancient, honourable, and substantial family, suffer any of the common earth of humanity to gather about its roots! The Rockvilles were so careful of their good blood, that they never allied it to any but blood as pure and inane as their own. Their elms flourished in the rotten earth of plebeian accumulations, and their acres produced large crops of corn from the sewage of towns and fat sinks, but the Rockvilles themselves took especial care that no vulgar vigour from the real heap of ordinary human nature should infuse a new force of intellect into their race. The Rockvilles needed nothing; they had all that an ancient, honourable, and substantial family could need. The Rockvilles had no need to study at school—why should they? They did not want to get on. The Rockvilles did not aspire to distinction for talent in the world—why should they? They had a large estate. So the Rockville soul, unused from generation to generation, grew—

Fine by degrees, and *spiritually* less,

till it tapered off into nothing.

Look at the last of a long line in the midst of his fine estate. Tall he was, with a stoop in his shoulders, and a bowing of his head on one side, as if he had been accustomed to stand under the low boughs of his woods, and peer after intruders. And that was precisely the fact. His features were thin and sharp; his nose prominent and keen in its character; his eyes small, black, and peering like a mole's, or a hungry swine's. Sir Roger was still oracular on the bench, after consulting his clerk, a good lawyer,—and looked up to by the neighbouring squires in election matters, for he was an unswerving tory. You never heard of a rational thing that he had said in the whole course of his life; but that mattered little, he was a gentleman of solemn aspect, of stately gait, and of a very ancient family.

With ten thousand a-year, and his rental rising, he was still, however, a man of overwhelming cares. What mattered a fine estate if all the world was against him? And Sir Roger firmly believed that he stood in that predicament. He had grown up to regard the world as full of little besides upstarts, radicals, manufacturers, and poachers. All were banded, in his belief, against the landed interest. It demanded all the energy of his very small faculties to defend himself and the world against them.

Unfortunately for his peace, a large manufacturing town had sprung up within a couple of miles of him. He could see its red-brick walls, and its red-tiled roofs, and its tall smoke-vomiting chimneys, growing and extending over the slopes beyond the river. It was to him the most irritating sight in the world; for what were all those swarming weavers and spinners but arrant radicals, upstarts, sworn foes of the ancient institutions and the landed interests of England? Sir Roger had passed through many a desperate conflict with them for the return of members to parliament. They brought forward men that were utter wormwood to all his feelings, and they paid no more respect to him and his friends on such occasions than they did to the meanest creature living. Reverence for ancient blood did not exist in that plebeian and rapidly multiplying tribe. There were master manufacturers there actually that looked and talked as big as himself, and *entre nous*, a vast deal more cleverly. The people talked of rights and franchises, and freedom of speech and of conscience, in a way that was really frightful. Then they were given most inveterately to running out in whole and everlasting crowds on Sundays and holidays into the fields and woods; and as there was no part of the neighbourhood half so pleasant as the groves and river banks of Rockville, they came swarming up there in crowds that were enough to drive any man of acres frantic.

Unluckily, there were roads all about Rockville; foot roads, and high roads, and bridle roads. There was a road up the river side, all the way to Rockville woods, and when it reached them, it divided like a fork, and one pony or footpath led straight up a magnificent grove of a mile long, ending close to the hall; and another ran all along the river side, under the hills and branches of the wood.

Oh, delicious were these woods! In the river there were islands, which were covered in summer with the greenest grass, and the freshest of willows, and the clear waters rushed around them in the most inviting manner imaginable. And there were numbers of people extremely ready to accept this delectable invitation of these waters. There they came in fine weather, and as these islands were only separated from the mainland by a little and very shallow stream, it was delightful for lovers to get across—with laughter, and treading on stepping-stones, and slipping off the stepping-stones up to the ankles into the cool brook, and pretty screams, and fresh laughter, and then landing on those sunny, and to them really enchanted, islands. And then came fishermen, solitary fishermen, and

fishermen in rows; fishermen lying in the flowery grass, with fragrant meadow-sweet and honey-breathing clover all about their ears; and fishermen standing in file, as if they were determined to clear all the river of fish in one day. And there were other lovers, and troops of loiterers, and shouting roysterers, going along under the boughs of the wood, and following the turns of that most companionable of rivers. And there were boats going up and down; boats full of young people, all holiday finery and mirth, and boats with duck-hunters and other, to Sir Roger, detestable marauders, with guns and dogs, and great bottles of beer. In the fine grove, on summer days, there might be found hundreds of people. There were picnic parties, fathers and mothers with whole families of children, and a grand promenade of the delighted artisans and their wives or sweethearts.

In the times prior to the sudden growth of the neighbouring town, Great Stockington, and to the simultaneous development of the love-of-nature principle in the Stockingtonians, nothing had been thought of all these roads. The roads were well enough till they led to these inroads. Then Sir Roger aroused himself. This must be changed. The roads must be stopped. Nothing was easier to his fancy. His fellow-justices, Sir Benjamin Bullockshed and Squire Sheepshank, had asked his aid to stop the like nuisances, and it had been done at once. So Sir Roger put up notices all about, that the roads were to be stopped by an Order of Session, and these notices were signed, as required by law, by their worships of Bullockshed and Sheepshank. But Sir Roger soon found that it was one thing to stop a road leading from One-man-Town to Lonely Lodge, and another to attempt to stop those from Great Stockington to Rockville.

On the very first Sunday after the exhibition of those notice-boards, there was a ferment in the grove of Rockville, as if all the bees in the county were swarming there, with all the wasps and hornets to boot. Great crowds were collected before each of these obnoxious placards, and the amount of curses vomited forth against them was really shocking for any day, but more especially for a Sunday. Presently there was a rush at them; they were torn down, and simultaneously pitched into the river. There were great crowds swarming all about Rockville all that day, and with looks so defiant that Sir Roger more than once contemplated sending off for the Yeoman Cavalry to defend his house, which he seriously thought in danger.

But so far from being intimidated from proceeding, this demonstration only made Sir Roger the more determined. To have so desperate and irreverent a population coming about his house and woods, now presented itself in a much more formidable aspect than ever. So, next day, not only were the placards once more hoisted, but rewards offered for the discovery of the offenders, attended with all the maledictions of the insulted majesty of the law. No notice was taken of this, but the whole of Great Stockington was in a buzz and an agitation. There were posters plastered all over the walls of the town, four times as large as Sir Roger's notices, in this style:—

“Englishmen! your dearest rights are menaced! The Woods of Rockville, your ancient, rightful, and enchanting resorts, are to be closed to you. Stockingtonians! the eyes of the world are upon you. ‘Awake! arise! or be for ever fallen!’ England expects every man to do his duty! And your duty is to resist and defy the grasping soil-lords, to seize on your ancient Patrimony!”

“Patrimony! Ancient and rightful resort of Rockville!” Sir Roger was astounded at the audacity of this upstart, plebeian race. What! they actually claimed Rockville, the heritage of a hundred successive Rockvilles, as their own. Sir Roger determined to carry it to the Sessions; and at the Sessions was a magnificent muster of all his friends. There was Sir Roger himself in the chair; and on either hand, a prodigious row of county squire-archy. There was Sir Benjamin Bullockshed, and Sir Thomas Tenterhook, and all the squires,—Sheepshank, Ramsbottom, Turnbull, Otterbrook, and Swagsides. The Clerk of Session read the notice for the closing of all the footpaths through the woods of Rockville, and declared that this notice had been duly, and for the required period publicly, posted. The Stockingtonians protested by their able lawyer Daredeville, against any order for the closing of these ancient woods—the inestimable property of the public.

“Property of the public!” exclaimed Sir Roger. “Property of the public!” echoed the multitudinous voices of indignant Bullocksheds, Tenterhooks, and Ramsbottoms. “Why, Sir, do you dispute the right of Sir Roger Rockville to his own estate?”

“By no means;” replied the undaunted Daredeville; “the estate of Rockville is unquestionably the property of the honourable baronet, Sir Roger Rockville; but the roads through it are the as unquestionable property of the public.”

The whole bench looked at itself; that is, at each other, in wrathful astonishment. The swelling in the diaphragms of the squires Otterbrook, Turnbull, and Swagsides, and all the rest of the worshipful row, was too big to admit of utterance. Only Sir Roger himself burst forth with an abrupt—

“Impudent fellows! But I’ll see them —— first!”

“Grant the order!” said Sir Benjamin Bullockshed; and the whole bench nodded assent. The able lawyer Daredeville retired with a pleasant smile. He saw an agreeable prospect of plenty of grist to his mill. Sir Roger was rich, and so was Great Stockington. He rubbed his hands, not in the least like a man defeated, and thought to himself, “Let them go at it—all right.”

The next day the placards on the Rockville estate were changed for others bearing “STOPPED BY ORDER OF SESSIONS!” and alongside of them were huge carefully painted boards, denouncing on all trespassers prosecutions according to law. The same evening came a prodigious invasion of Stockingtonians—tore all the boards and placards down, and carried them on their shoulders to Great Stockington, singing as they went, “See, the Conquering Heroes come!” They set them up in the centre of the Stockington marketplace, and burnt them, along with, an effigy of Sir Roger Rockville.

That was grist at once to the mill of the able lawyer Daredeville. He looked on, and rubbed his hands. Warrants were speedily issued by the Baronets of Bullockshed and Tenterhook, for the apprehension of the individuals who had been seen carrying off the notice-boards, for larceny, and against a number of others for trespass. There was plenty of work for Daredeville and his brethren of the robe; but it all ended, after the flying about of sundry mandamuses and assize trials, in Sir Roger finding that though Rockville was his, the roads through it were the public’s.

As Sir Roger drove homeward from the assize, which finally settled the question of these footpaths, he heard the bells in all the steeples of Great Stockington burst forth with a grand peal of triumph. He closed first the windows of his fine old carriage, and sunk into a corner; but he could not drown the intolerable sound. “But,” said he, “I’ll stop their pic-nic-ing. I’ll stop their fishing. I’ll have hold of them for trespassing and poaching!” There was war henceforth between Rockville and Great Stockington.

On the very next Sunday there came literally thousands of the jubilant Stockingtonians to Rockville. They had brought baskets, and were for

dining, and drinking success to all footpaths. But in the great grove there were keepers, and watchers, who warned them to keep the path, that narrow well-worn line up the middle of the grove. “What! were they not to sit on the grass?”—“No!”—“What! were they not to pic-nic?”—“No! not there!”

The Stockingtonians felt a sudden damp on their spirits. But the river bank! The cry was “To the river bank! There they *would* pic-nic.” The crowd rushed away down the wood, but on the river bank they found a whole regiment of watchers, who pointed again to the narrow line of footpath, and told them not to trespass beyond it. But the islands! they went over to the islands. But there too were Sir Roger’s forces, who warned them back! There was no road there—all found there would be trespassers, and be duly punished.

The Stockingtonians discovered that their triumph was not quite so complete as they had flattered themselves. The footpaths were theirs, but that was all. Their ancient license was at an end. If they came there, there was no more fishing; if they came in crowds, there was no more pic-nic-ing; if they walked through the woods in numbers, they must keep to Indian file, or they were summoned before the county magistrates for trespass, and were soundly fined; and not even the able Daredeville would undertake to defend them.

The Stockingtonians were chop-fallen, but they were angry and dogged; and they thronged up to the village and the front of the hall. They filled the little inn in the hamlet—they went by scores, and roving all over the churchyard, read epitaphs

That teach the rustic moralists to die,

but don’t teach them to give up their old indulgences very good-humouredly. They went and sat in rows on the old churchyard wall, opposite to the very windows of the irate Sir Roger. They felt themselves beaten, and Sir Roger felt himself beaten. True, he could coerce them to the keeping of the footpaths—but, then, they had the footpaths! True, thought the Stockingtonians, we have the footpaths, but then the pic-nic-ing, and the fishing, and the islands! The Stockingtonians were full of sullen wrath, and Sir Roger was—oh, most expressive old Saxon phrase—*HAIRSORE*! Yes, he was one universal round of vexation and jealousy of his rights. Every hair in his body was like a pin sticking into him. Come within a dozen yards of

him; nay, at the most, blow on him, and he was excruciated—you rubbed his sensitive hairs at a furlong's distance.

The next Sunday the people found the churchyard locked up, except during service, when beadles walked there, and desired them not to loiter and disturb the congregation, closing the gates, and showing them out like a flock of sheep the moment the service was over. This was fuel to the already boiling blood of Stockington. The week following, what was their astonishment to find a much frequented ruin gone! it was actually gone! not a trace of it; but the spot where it had stood for ages, turfed, planted with young spruce trees, and fenced off with post and rail! The exasperated people now launched forth an immensity of fulminations against the churl Sir Roger, and a certain number of them resolved to come and seat themselves in the street of the hamlet and there dine; but a terrific thunderstorm, which seemed in league with Sir Roger, soon routed them, drenched them through, and on attempting to seek shelter in the cottages, the poor people said they were very sorry, but it was as much as their holdings were worth, and they dare not admit them.

Sir Roger had triumphed! It was all over with the old delightful days at Rockville. There was an end of pic-nic-ing, of fishing, and of roving in the islands. One sturdy disciple of Izaak Walton, indeed, dared to fling a line from the banks of Rockville grove, but Sir Roger came upon him and endeavoured to seize him. The man coolly walked into the middle of the river, and, without a word, continued his fishing.

“Get out there!” exclaimed Sir Roger, “that is still on my property.” The man walked through the river to the other bank, where he knew that the land was rented by a farmer. “Give over,” shouted Sir Roger, “I tell you the water is mine.”

“Then,” said the fellow, “bottle it up, and be hanged to you! Don’t you see it is running away to Stockington?”

There was bad blood between Rockville and Stockington-green. Stockington was incensed, and Sir Roger was hairsore.

A new nuisance sprung up. The people of Stockington looked on the cottagers of Rockville as sunk in deepest darkness under such a man as Sir Roger and his cousin the vicar. They could not pic-nic, but they thought they could hold a camp-meeting; they could not fish for roach, but they thought they might for souls. Accordingly there assembled crowds of

Stockingtonians on the green of Rockville, with a chair and a table, and a preacher with his head bound in a red handkerchief; and soon there was a sound of hymns, and a zealous call to come out of the darkness of the spiritual Babylon. But this was more than Sir Roger could bear; he rushed forth with all his servants, keepers, and cottagers, overthrew the table, and routing the assembly, chased them to the boundary of his estate.

The discomfited Stockingtonians now fulminated awful judgments on the unhappy Sir Roger, as a persecutor and a malignant. They dared not enter again on his park, but they came to the very verge of it, and held weekly meetings on the highway, in which they sang and declaimed as loudly as possible, that the winds might bear their voices to Sir Roger's ears.

To such a position was now reduced the last of the long line of Rockville. The spirit of a policeman had taken possession of him. He had keepers and watchers out on all sides, but that did not satisfy him. He was perpetually haunted with the idea that poachers were after his game, that trespassers were in his woods. His whole life was now spent in stealing to and fro in his fields and plantations, and prowling along his river side. He looked under hedges, and watched for long hours under the forest trees. If any one had a curiosity to see Sir Roger, they had only to enter his fields by the wood side, and wander a few yards from the path, and he was almost sure to spring out over the hedge, and in angry tones demand their name and address. The descendant of the chivalrous and steel-clad De Rockvilles was sunk into a restless spy on his own ample property. There was but one idea in his mind—encroachment. It was destitute of all other furniture but the musty technicalities of warrants and commitments. There was a stealthy and skulking manner in everything that he did. He went to church on Sundays, but it was no longer by the grand iron gate opposite to his house, that stood generally with a large spider's web woven over the lock, and several others in different corners of the fine iron tracery, bearing evidence of the long period since it had been opened. How different to the time when the Sir Roger and Lady of Rockville had had these gates thrown wide on a Sunday morning, and, with all their train of household servants after their back, with true antique dignity, marched with much proud humility into the house of God. Now, Sir Roger—the solitary, suspicious, undignified Sir Roger, the keeper and policeman of his own property—stole in at a little side gate from his paddock, and back the same way, wondering all the time whether there

was not somebody in his pheasant preserves, or Sunday trespassers in his grove.

If you entered his house, it gave you as cheerless a feeling as its owner. There was the conservatory, so splendid with rich plants and flowers in his mother's time—now a dusty receptacle of hampers, broken hand-glasses, and garden tools. These tools could never be used, for the gardens were grown wild. Tall grass grew in the walks, and the huge unpruned shrubs disputed the passage with you. In the wood above the gardens, reached by several flights of fine, but now moss-grown, steps, there stood a pavilion, once clearly very beautiful. It was now damp and ruinous—its walls covered with greenness and crawling insects. It was a great lurking-place of Sir Roger when on the watch for poachers.

The line of the Rockvilles was evidently running fast out. It had reached the extremity of imbecility and contempt—it must soon reach its close.

Sir Roger used to make his regular annual visit to town; but of late, when there, he had wandered restlessly about the streets, peeping into the shop-windows; and if it rained, standing under entries for hours after, till it was gone over. The habit of lurking and peering about, was upon him; and his feet bore him instinctively into those narrow and crowded alleys where swarm the poachers of the city—the trespassers and anglers in the game preserves and streams of humanity. He had lost all pleasure in his club; the most exciting themes of political life retained no piquancy for him. His old friends ceased to find any pleasure in him. He was become the driest of all dry wells. Poachers, and anglers, and Methodists, haunted the wretched purlieus of his lost fading-out mind, and he resolved to go to town no more. His whole nature was centred in his woods. He was for ever on the watch; and when at Rockville again, if he heard a door clap when in bed, he thought it a gun in his woods, and started up, and was out with his keepers.

Of what value was that magnificent estate to him?—those superb woods; those finely-hanging cliffs; that clear and *riant* river coming travelling on, and taking a noble sweep below his windows,—that glorious expanse of neat verdant meadows stretching almost to Stockington, and enlivened by numerous herds of the most beautiful cattle—those old farms and shady lanes overhung with hazel and wild rose; the glittering brook, and the songs of woodland birds—what were they to that worn-out old man, that victim of the delusive doctrine of blood, of the man-trap of an hereditary name?

There the poet could come, and feel the presence of divinity in that noble scene, and hear sublime whispers in the trees, and create new heavens and earths from the glorious chaos of nature around him, and in one short hour live an empyrean of celestial life and love. There could come the very humblest children of the plebeian town, and feel a throb of exquisite delight pervade their bosoms at the sight of the very flowers on the sod, and see heaven in the infinite blue above them. And poor Sir Roger, the holder, but not the possessor of all, walked only in a region of sterility, with no sublimer ideas than poachers and trespassers—no more rational enjoyment than the brute indulgence of hunting like a ferret, and seizing his fellow-men like a bulldog. He was a specimen of human nature degenerated, retrograded from the divine to the bestial, through the long-operating influences of false notions and institutions, continued beyond their time. He had only the soul of a keeper. Had he been only a keeper, he had been a much happier man.

His time was at hand. The severity which he had long dealt out towards all sorts of offenders made him the object of the deepest vengeance. In a lonely hollow of his woods, watching at midnight with two of his men, there came a sturdy knot of poachers. An affray ensued. The men perceived that their old enemy, Sir Roger, was there: and the blow of a hedge-stake stretched him on the earth. His keepers fled—and thus ignominiously terminated the long line of the Rockvilles. Sir Roger was the last of his line, but not of his class. There is a feudal art of sinking, which requires no study; and the Rockvilles are but one family amongst thousands who have perished in its practice.

THE CHEMISTRY OF A CANDLE.

The Wilkinsons were having a small party,—it consisted of themselves and Uncle Bagges—at which the younger members of the family, home for the holidays, had been just admitted to assist after dinner. Uncle Bagges was a gentleman from whom his affectionate relatives cherished expectations of a testamentary nature. Hence the greatest attention was paid by them to the wishes of Mr. Bagges, as well as to every observation which he might be pleased to make.

“Eh! what? you sir,” said Mr. Bagges, facetiously addressing himself to his eldest nephew, Harry,—“Eh! what? I am glad to hear, sir, that you are doing well at school. Now—eh? now, are you clever enough to tell me where was Moses when he put the candle out?”

“That depends, uncle,” answered the young gentleman, “on whether he had lighted the candle to see with at night, or by daylight, to seal a letter.”

“Eh! Very good, now! ’Pon my word, very good,” exclaimed Uncle Bagges. “You must be Lord Chancellor, sir—Lord Chancellor, one of these days.”

“And now, uncle,” asked Harry, who was a favourite with the old gentleman, “can you tell me what you do when you put a candle out?”

“Clap an extinguisher on it, you young rogue, to be sure.”

“Oh! but I mean, you cut off its supply of oxygen,” said Master Harry.

“Cut off its ox’s—eh? what? I shall cut off your nose, you young dog, one of these fine days.”

“He means something he heard at the Royal Institution,” observed Mrs. Wilkinson. “He reads a great deal about chemistry, and he attended Professor Faraday’s lectures there on the chemical history of a candle, and has been full of it ever since.”

“Now, you sir,” said Uncle Bagges, “come you here to me, and tell me what you have to say about this chemical, eh?—or comical; which?—this comical chemical history of a candle.”

“He’ll bore you, Bagges,” said Mr. Wilkinson. “Harry, don’t be troublesome to your uncle.”

“Troublesome! Oh, not at all. He amuses me. I like to hear him. So let him teach his old uncle the comicality and chemicality of a farthing rushlight.”

“A wax candle will be nicer and cleaner, uncle, and answer the same purpose. There’s one on the mantel-shelf. Let me light it.”

“Take care you don’t burn your fingers, or set anything on fire,” said Mrs. Wilkinson.

“Now, uncle,” commenced Harry, having drawn his chair to the side of Mr. Bagges, “we have got our candle burning. What do you see?”

“Let me put on my spectacles,” answered the uncle.

“Look down on the top of the candle around the wick. See, it is a little cup full of melted wax. The heat of the flame has melted the wax just round the wick. The cold air keeps the outside of it hard, so as to make the rim of it. The melted wax in the little cup goes up through the wick to be burnt, just as oil does in the wick of a lamp. What do you think makes it go up, uncle?”

“Why—why, the flame draws it up, doesn’t it?”

“Not exactly, uncle. It goes up through little tiny passages in the cotton wick, because very, very small channels, or pipes, or pores, have the power in themselves of sucking up liquids. What they do it by is called cap—something.”

“Capillary attraction, Harry,” suggested Mr. Wilkinson.

“Yes, that’s it; just as a sponge sucks up water, or a bit of lump-sugar the little drop of tea or coffee left in the bottom of a cup. But I mustn’t say much more about this, or else you will tell me I am doing something very much like teaching my grandmother to—you know what.”

“Your grandmother, eh, young sharpshins?”

“No—I mean my uncle. Now, I’ll blow the candle out, like Moses; not to be in the dark, though, but to see into what it is. Look at the smoke rising

from the wick. I'll hold a bit of lighted paper in the smoke, so as not to touch the wick. But see, for all that, the candle lights again. So this shows that the melted wax sucked up through the wick is turned into vapour; and the vapour burns. The heat of the burning vapour keeps on melting more wax, and that is sucked up too within the flame, and turned into vapour, and burnt, and so on till the wax is all used up, and the candle is gone. So the flame, uncle, you see, is the last of the candle, and the candle seems to go through the flame into nothing—although it doesn't, but goes into several things, and isn't it curious, as Professor Faraday said, that the candle should look so splendid and glorious in going away?"

"How well he remembers, doesn't he?" observed Mrs. Wilkinson.

"I dare say," proceeded Harry, "that the flame of the candle looks flat to you; but if we were to put a lamp glass over it, so as to shelter it from the draught, you would see it is round,—round sideways, and running up to a peak. It is drawn up by the hot air; you know that hot air always rises, and that is the way smoke is taken up the chimney. What should you think was in the middle of the flame?"

"I should say, fire," replied Uncle Bagges.

"Oh, no! The flame is hollow. The bright flame we see is something no thicker than a thin peel, or skin; and it doesn't touch the wick. Inside of it is the vapour I told you of just now. If you put one end of a bent pipe into the middle of the flame, and let the other end of the pipe dip into a bottle, the vapour or gas from the candle will mix with the air there; and if you set fire to the mixture of gas from the candle and air in the bottle, it would go off with a bang."

"I wish you'd do that, Harry," said Master Tom, the younger brother of the juvenile lecturer.

"I want the proper things," answered Harry. "Well, uncle, the flame of the candle is a little shining case, with gas in the inside of it, and air on the outside, so that the case of flame is between the air and the gas. The gas keeps going into the flame to burn, and when the candle burns properly, none of it ever passes out through the flame; and none of the air ever gets in through the flame to the gas. The greatest heat of the candle is in this skin, or peel, or case of flame."

"Case of flame!" repeated Mr. Bagges. "Live and learn. I should have thought a candle-flame was as thick as my poor old noddle."

“I can show you the contrary,” said Harry. “I take this piece of white paper, look, and hold it a second or two down upon the candle-flame, keeping the flame very steady. Now I’ll rub off the black of the smoke, and—there—you find that the paper is scorched in the shape of a ring; but inside the ring it is only dirtied, and not singed at all.”

“Seeing is believing,” remarked the uncle.

“But,” proceeded Harry, “there is more in the candle-flame than the gas that comes out of the candle. You know a candle won’t burn without air. There must be always air around the gas, and touching it like, to make it burn. If a candle hasn’t got enough air, it goes out, or burns badly, so that some of the vapour inside of the flame comes out through it in the form of smoke, and this is the reason of a candle smoking. So now you know why a great clumsy dip smokes more than a neat wax candle; it is because the thick wick of the dip makes too much fuel in proportion to the air that can get to it.”

“Dear me! Well, I suppose there is a reason for everything,” exclaimed the young philosopher’s mamma.

“What should you say, now,” continued Harry, “if I told you that the smoke that comes out of a candle is the very thing that makes a candle light? Yes; a candle shines by consuming its own smoke. The smoke of a candle is a cloud of small dust, and the little grains of the dust are bits of charcoal, or carbon, as chemists call it. They are made in the flame, and burnt in the flame, and, while burning, make the flame bright. They are burnt the moment they are made; but the flame goes on making more of them as fast as it burns them; and that is how it keeps bright. The place they are made in, is in the case of flame itself, where the strongest heat is. The great heat separates them from the gas which comes from the melted wax, and, as soon as they touch the air on the outside of the thin case of flame, they burn.”

“Can you tell how it is that the little bits of carbon cause the brightness of the flame?” asked Mr. Wilkinson.

“Because they are pieces of solid matter,” answered Harry. “To make a flame shine, there must always be some solid—or at least liquid—matter in it.”

“Very good,” said Mr. Bagges,—“solid stuff necessary to brightness.”

“Some gases and other things,” resumed Harry, “that burn with a flame you can hardly see, burn splendidly when something solid is put into them. Oxygen and hydrogen—tell me if I use too hard words, uncle—oxygen and hydrogen gases, if mixed together and blown through a pipe, burn with plenty of heat but with very little light. But if their flame is blown upon a piece of quicklime, it gets so bright as to be quite dazzling. Make the smoke of oil of turpentine pass through the same flame, and it gives the flame a beautiful brightness directly.”

“I wonder,” observed Uncle Bagges, “what has made you such a bright youth.”

“Taking after uncle, perhaps,” retorted his nephew. “Don’t put my candle and me out. Well, carbon or charcoal is what causes the brightness of all lamps, and candles, and other common lights; so, of course, there is carbon in what they are all made of.”

“So carbon is smoke, eh? and light is owing to your carbon. Giving light out of smoke, eh? as they say in the classics,” observed Mr. Bagges.

“But what becomes of the candle,” pursued Harry, “as it burns away? where does it go?”

“Nowhere,” said his mamma, “I should think. It burns to nothing.”

“Oh, dear, no!” said Harry, “everything—everybody goes somewhere.”

“Eh!—rather an important consideration that,” Mr. Bagges moralised.

“You can see it goes into smoke, which makes soot for one thing,” pursued Harry. “There are other things it goes into, not to be seen by only looking, but you can get to see them by taking the right means,—just put your hand over the candle, uncle.”

“Thank you, young gentleman, I had rather be excused.”

“Not close enough down to burn you, uncle; higher up. There,—you feel a stream of hot air; so something seems to rise from the candle. Suppose you were to put a very long slender gas-burner over the flame, and let the flame burn just within the end of it, as if it were a chimney,—some of the hot steam would go up and come out at the top, but a sort of dew would be left behind in the glass chimney, if the chimney was cold enough when you put it on. There are ways of collecting this sort of dew, and when it is collected it turns out to be really water. I am not joking, uncle. Water is one of the things which the candle turns into in burning,—water coming out of

fire. A jet of oil gives above a pint of water in burning. In some lighthouses they burn, Professor Faraday says, up to two gallons of oil in a night, and if the windows are cold the steam from the oil clouds the inside of the windows, and, in frosty weather, freezes into ice.”

“Water out of a candle, eh?” exclaimed Mr. Bagges. “As hard to get, I should have thought, as blood out of a post. Where does it come from?”

“Part from the wax, and part from the air, and yet not a drop of it comes either from the air or the wax. What do you make of that, uncle?”

“Eh? Oh! I’m no hand at riddles. Give it up.”

“No riddle at all, uncle. The part that comes from the wax isn’t water, and the part that comes from the air isn’t water, but when put together they become water. Water is a mixture of two things, then. This can be shown. Put some iron wire or turnings into a gun-barrel open at both ends. Heat the middle of the barrel red-hot in a little furnace. Keep the heat up, and send the steam of boiling water through the red-hot gun-barrel. What will come out at the other end of the barrel won’t be steam; it will be gas, which doesn’t turn to water again when it gets cold, and which burns if you put a light to it. Take the turnings out of the gun-barrel, and you will find them changed to rust, and heavier than when they were put in. Part of the water is the gas that comes out of the barrel, the other part is what mixes with the iron turnings, and changes them to rust, and makes them heavier. You can fill a bladder with the gas that comes out of the gun-barrel, or you can pass bubbles of it up into a jar of water turned upside down in a trough, and, as I said, you can make this part of the water burn.”

“Eh?” cried Mr. Bagges. “Upon my word! One of these days, we shall have you setting the Thames on fire.”

“Nothing more easy,” said Harry, “than to burn part of the Thames, or of any other water; I mean the gas that I have just told you about, which is called hydrogen. In burning, hydrogen produces water again, like the flame of the candle. Indeed, hydrogen is that part of the water, formed by a candle burning, that comes from the wax. All things that have hydrogen in them produce water in burning, and the more there is in them the more they produce. When pure hydrogen burns, nothing comes from it but water, no smoke or soot at all. If you were to burn one ounce of it, the water you would get would be just nine ounces. There are many ways of making hydrogen, besides out of steam by the hot gun-barrel. I could show it you in

a moment by pouring a little sulphuric acid mixed with water into a bottle upon a few zinc or steel filings, and putting a cork in the bottle with a little pipe through it, and setting fire to the gas that would come from the mouth of the pipe. We should find the flame very hot, but having scarcely any brightness. I should like you to see the curious qualities of hydrogen, particularly how light it is, so as to carry things up in the air; and I wish I had a small balloon to fill with it and make go up to the ceiling, or a bag-pipe full of it to blow soap-bubbles with, and show how much faster they rise than common ones, blown with the breath."

"So do I," interposed Master Tom.

"And so," resumed Harry, "hydrogen, you know, uncle, is part of water, and just one-ninth part."

"As hydrogen is to water, so is a tailor to an ordinary individual, eh?" Mr. Bagges remarked.

"Well, now then, uncle, if hydrogen is the tailor's part of the water, what are the other eight parts? The iron turnings used to make hydrogen in the gun-barrel, and rusted, take just those eight parts from the water in the shape of steam, and are so much the heavier. Burn iron turnings in the air, and they make the same rust, and gain just the same in weight. So the other eight parts must be found in the air for one thing, and in the rusted iron turnings for another, and they must also be in the water; and now the question is, how to get at them?"

"Out of the water? Fish for them, I should say," suggested Mr. Bagges.

"Why, so we can," said Harry. "Only, instead of hooks and lines, we must use wires—two wires, one from one end, the other from the other, of a galvanic battery. Put the points of these wires into water, a little distance apart, and they instantly take the water to pieces. If they are of copper, or a metal that will rust easily, one of them begins to rust, and air-bubbles come up from the other. These bubbles are hydrogen. The other part of the water mixes with the end of the wire and makes rust. But if the wires are of gold, or a metal that does not rust easily, air-bubbles rise from the ends of both wires. Collect the bubbles from both wires in a tube, and fire them, and they turn to water again; and this water is exactly the same weight as the quantity that has been changed into the two gases. Now then, uncle, what should you think water was composed of?"

“Eh? well—I suppose of those very identical two gases, young gentleman.”

“Right, uncle. Recollect that the gas from one of the wires was hydrogen, the one-ninth of water. What should you guess the gas from the other wire to be?”

“Stop—eh?—wait a bit—eh?—oh!—why, the other eight-ninths, to be sure.”

“Good again, uncle. Now this gas that is eight-ninths of water is the gas called oxygen that I mentioned just now. This is a very curious gas. It won’t burn in air at all itself, like gas from a lamp, but it has a wonderful power of making things burn that are lighted and put into it. If you fill a jar with it _____”

“How do you manage that?” Mr. Bagges inquired.

“You fill the jar with water,” answered Harry, “and you stand it upside down in a vessel full of water too. Then you let bubbles of the gas up into the jar and they turn out the water and take its place. Put a stopper in the neck of the jar, or hold a glass plate against the mouth of it, and you can take it out of the water and so have bottled oxygen. A lighted candle put into a jar of oxygen blazes up directly and is consumed before you can say Jack Robinson. Charcoal burns away in it as fast, with beautiful bright sparks—phosphorus with a light that dazzles you to look at—and a piece of iron or steel just made red-hot at the end first, is burnt in oxygen quicker than a stick would be in common air. The experiment of burning things in oxygen beats any fire-works.”

“Oh, how jolly!” exclaimed Tom.

“Now we see, uncle,” Harry continued, “that water is hydrogen and oxygen united together, that water is got wherever hydrogen is burnt in common air, that a candle won’t burn without air, and that when a candle burns there is hydrogen in it burning, and forming water. Now, then, where does the hydrogen of the candle get the oxygen from, to turn into water with it?”

“From the air, eh?”

“Just so. I can’t stop to tell you of the other things which there is oxygen in, and the many beautiful and amusing ways of getting it. But as there is oxygen in the air, and as oxygen makes things burn at such a rate, perhaps

you wonder why air does not make things burn as fast as oxygen. The reason is, that there is something else in the air that mixes with the oxygen and weakens it.”

“Makes a sort of gaseous grog of it, eh?” said Mr. Bagges. “But how is that proved?”

“Why, there is a gas, called nitrous gas, which, if you mix it with oxygen, takes all the oxygen into itself, and the mixture of the nitrous gas and oxygen, if you put water with it, goes into the water. Mix nitrous gas and air together in a jar over water, and the nitrous gas takes away the oxygen, and then the water sucks up the mixed oxygen and nitrous gas, and that part of the air which weakens the oxygen is left behind. Burning phosphorus in confined air will also take all the oxygen from it, and there are other ways of doing the same thing. The portion of the air left behind is called nitrogen. You wouldn’t know it from common air by the look; it has no colour, taste, nor smell, and it won’t burn. But things won’t burn in it, either; and anything on fire put into it goes out directly. It isn’t fit to breathe,—and a mouse, or any animal, shut up in it, dies. It isn’t poisonous, though; creatures only die in it for want of oxygen. We breathe it with oxygen, and then it does no harm, but good; for if we breathed pure oxygen, we should breathe away so violently, that we should soon breathe our life out. In the same way, if the air were nothing but oxygen, a candle would not last above a minute.”

“What a tallow-chandler’s bill we should have!” remarked Mrs. Wilkinson.

“‘If a house were on fire in oxygen,’ as Professor Faraday said, ‘every iron bar, or rafter, or pillar, every nail and iron tool, and the fire-place itself; all the zinc and copper roofs, and leaden coverings, and gutters, and pipes, would consume and burn, increasing the combustion.’”

“That would be, indeed, burning ‘like a house on fire,’” observed Mr. Bagges.

“‘Think,’” said Harry, continuing his quotation, “‘of the Houses of Parliament, or a steam-engine manufactory. Think of an iron-proof chest—no proof against oxygen. Think of a locomotive and its train,—every engine, every carriage, and even every rail would be set on fire and burnt up.’ So now, uncle, I think you see what the use of nitrogen is, and especially how it prevents a candle from burning out too fast.”

“Eh?” said Mr. Bagges. “Well, I will say I do think we are under considerable obligations to nitrogen.”

“I have explained to you, uncle,” pursued Harry, “how a candle, in burning, turns into water. But it turns into something else besides that; there is a stream of hot air going up from it that won’t condense into dew; some of that is the nitrogen of the air which the candle has taken all the oxygen from. But there is more in it than nitrogen. Hold a long glass tube over a candle, so that the stream of hot air from it may go up through the tube. Hold a jar over the end of the tube to collect some of the stream of hot air. Put some lime-water, which looks quite clear, into the jar; stop the jar, and shake it up. The lime-water, which was quite clear before, turns milky. Then there is something made by the burning of the candle that changes the colour of the lime-water. That is a gas, too, and you can collect it, and examine it. It is to be got from several things, and is a part of all chalk, marble, and the shells of eggs or of shell-fish. The easiest way to make it is by pouring muriatic or sulphuric acid on chalk or marble. The marble or chalk begins to hiss or bubble, and you can collect the bubbles in the same way that you can oxygen. The gas made by the candle in burning, and which also is got out of the chalk and marble, is called carbonic acid. It puts out a light in a moment; it kills any animal that breathes it, and it is really poisonous to breathe, because it destroys life even when mixed with a pretty large quantity of common air. The bubbles made by beer when it ferments, are carbonic acid, so is the air that fizzes out of soda-water,—and it is good to swallow though it is deadly to breathe. It is got from chalk by burning the chalk as well as by putting acid to it, and burning the carbonic acid out of chalk makes the chalk lime. This is why people are killed sometimes by getting in the way of the wind that blows from lime-kilns.”

“Of which it is advisable carefully to keep to the windward,” Mr. Wilkinson observed.

“The most curious thing about carbonic acid gas,” proceeded Harry, “is its weight. Although it is only a sort of air, it is so heavy that you can pour it from one vessel into another. You may dip a cup of it and pour it down upon a candle, and it will put the candle out, which would astonish an ignorant person; because carbonic acid gas is as invisible as the air, and the candle seems to be put out by nothing. A soap-bubble or common air floats on it like wood on water. Its weight is what makes it collect in brewers’ vats; and

also in wells, where it is produced naturally; and owing to its collecting in such places it causes the deaths we so often hear about of those who go down into them without proper care. It is found in many springs of water, more or less; and a great deal of it comes out of the earth in some places. Carbonic acid gas is what stupifies the dogs in the Grotto del Cane. Well, but how is carbonic acid gas made by the candle?"

"I hope with your candle you you'll throw some light upon the subject," said Uncle Bagges.

"I hope so," answered Harry. "Recollect it is the burning of the smoke, or soot, or carbon of the candle that makes the candle-flame bright. Also that the candle won't burn without air. Likewise that it will not burn in nitrogen, or air that has been deprived of oxygen. So the carbon of the candle mingles with oxygen, in burning, to make carbonic acid gas, just as the hydrogen does to form water. Carbonic acid gas, then, is carbon or charcoal dissolved in oxygen. Here is black soot getting invisible and changing into air; and this seems strange, uncle, doesn't it?"

"Ahem! Strange, if true," answered Mr. Bagges. "Eh?—well! I suppose it's all right."

"Quite so, uncle. Burn carbon or charcoal either in the air or in oxygen, and it is sure always to make carbonic acid, and nothing else, if it is dry. No dew or mist gathers in a cold glass jar if you burn dry charcoal in it. The charcoal goes entirely into carbonic acid gas, and leaves nothing behind but ashes, which are only earthy stuff that was in the charcoal, but not part of the charcoal itself. And now, shall I tell you something about carbon?"

"With all my heart," assented Mr. Bagges.

"I said that there was carbon or charcoal in all common lights,—so there is in every common kind of fuel. If you heat coal or wood away from the air, some gas comes away, and leaves behind coke from coal, and charcoal from wood; both carbon, though not pure. Heat carbon as much as you will in a close vessel, and it does not change in the least; but let the air get to it, and then it burns and flies off in carbonic acid gas. This makes carbon so convenient for fuel. But it is ornamental as well as useful, uncle. The diamond is nothing else than carbon."

"The diamond, eh? You mean the black diamond."

“No; the diamond, really and truly. The diamond is only carbon in the shape of a crystal.”

“Eh? and can’t some of your clever chemists crystallise a little bit of carbon, and make a Koh-i-noor?”

“Ah, uncle, perhaps we shall, some day. In the meantime I suppose we must be content with making carbon so brilliant as it is in the flame of a candle. Well; now you see that a candle-flame is vapour burning, and the vapour, in burning, turns into water and carbonic acid gas. The oxygen of both the carbonic acid gas and the water comes from the air, and the hydrogen and carbon together are the vapour. They are distilled out of the melted wax by the heat. But, you know, carbon alone can’t be distilled by any heat. It can be distilled, though, when it is joined with hydrogen, as it is in the wax, and then the mixed hydrogen and carbon rise in gas of the same kind as the gas in the streets, and that also is distilled by heat from coal. So a candle is a little gas manufactory in itself, that burns the gas as fast as it makes it.”

“Haven’t you pretty nearly come to your candle’s end?” said Mr. Wilkinson.

“Nearly. I only want to tell uncle, that the burning of a candle is almost exactly like our breathing. Breathing is consuming oxygen, only not so fast as burning. In breathing we throw out water in vapour and carbonic acid from our lungs, and take oxygen in. Oxygen is as necessary to support the life of the body, as it is to keep up the flame of a candle.”

“So,” said Mr. Bagges, “man is a candle, eh? and Shakespeare knew that, I suppose, (as he did most things,) when he wrote—

‘Out, out, brief candle!’

“Well, well; we old ones are moulds, and you young squires are dips and rushlights, eh? Any more to tell us about the candle?”

“I could tell you a great deal more about oxygen, and hydrogen, and carbon, and water, and breathing, that Professor Faraday said, if I had time; but you should go and hear him yourself, uncle.”

“Eh? well! I think I will. Some of us seniors may learn something from a juvenile lecture, at any rate, if given by a Faraday. And now, my boy, I will tell you what,” added Mr. Bagges, “I am very glad to find you so fond of study and science; and you deserve to be encouraged; and so I’ll give you a

what-d' ye-call-it?—a Galvanic Battery on your next birth-day; and so much for your teaching your old uncle the chemistry of a candle.”

AN OLD HAUNT.

The rippling water, with its drowsy tone,—
The tall elms, tow'ring in their stately pride,—
And—sorrow's type—the willow sad and lone,
Kissing in graceful woe the murmuring tide;—

The grey church-tower,—and dimly seen beyond,
The faint hills gilded by the parting sun,—
All were the same, and seem'd with greeting fond
To welcome me as they of old had done.

And for a while I stood as in a trance,
On that loved spot, forgetting toil and pain;—
Buoyant my limbs, and keen and bright my glance,
For that brief space I was a boy again!

Again with giddy mates I careless play'd,
Or plied the quiv'ring oar, on conquest bent;—
Again, beneath the tall elms' silent shade,
I woo'd the fair, and won the sweet consent.

But brief, alas! the spell,—for suddenly
Peal'd from the tower the old familiar chimes,
And with their clear, heart-thrilling melody,
Awaked the spectral forms of darker times.

And I remember'd all that years had wrought—
How bow'd my care-worn frame, how dimm'd my eye,
How poor the gauds by Youth so keenly sought,
How quench'd and dull Youth's aspirations high!

And in half mournful, half upbraiding host,
Duties neglected—high resolves unkept—
And many a heart by death or falsehood lost,
In lightning current o'er my bosom swept.

Then bow'd the stubborn knees, as backward sped
The self-accusing thoughts in dread array,
And slowly, from their long-congealèd bed,
Forced the remorseful tears their silent way.

Bitter, yet healing drops! in mercy sent,
Like soft dews falling on a thirsty plain,—
And 'ere those chimes their last faint notes had spent,
Strengthen'd and calm'd, I stood erect again.

Strengthen'd, the tasks allotted to fulfil;—
Calm'd, the thick-coming sorrows to endure;
Fearful of nought but of my own frail will,—
In His Almighty strength and aid secure.

For a sweet voice had whisper'd hope to me,—
Had through my darkness shed a kindly ray;—
It said: "The past is fix'd immutably,
Yet is there comfort in the coming day!"

THE HIPPOPOTAMUS.

Before we give a more exclusive attention to the “illustrious stranger,” we think it will be advisable to present the reader with a brief authentic account of the circumstances which led to the honour conferred upon England by the visit of this extraordinary personage. These circumstances are little known to the world; indeed, we have reason to believe they have never before been published.

The British Consul at Cairo had frequently intimated to His Highness the Pasha of Egypt, that a live hippopotamus would be regarded as a very interesting and valuable present in England. Now, there were sundry difficulties of a serious nature involved in this business. In the first place, the favourite resort of the hippopotami is a thousand or fifteen hundred miles distant from Cairo; in the second place, the hippopotamus being amphibious, is not easily come-at-able; when he is environed, he is a tremendous antagonist, by reason of his great strength, enormous weight, his wrathfulness when excited, and we may add his prodigious mouth with its huge tusks. We are speaking of the *male* hippopotamus. He is often slain by a number of rifle-balls (he only makes a comic grin of scorn at a few) and laid low from a distance: but as to being taken alive, that is a triumph which has scarcely ever been permitted to mortal man of modern times. It is quite a different matter in respect of the elephant. He cannot take to the water, and neither dive clean away, nor upset your boat with a plunge of his forehead; besides which you cannot get two tame renegade hippopotami to assist in the capture and subjugation of a relative, as is the case with elephants. Accordingly, His Highness the Pasha, not liking to compromise the dignity of despotism, and his own position as sovereign of Egypt, by promising anything which he might, perhaps, be unable to perform, turned a deaf ear to the repeated overtures of the British Consul. He never refused

his request; he simply did not hear what he said, or could not be made to have a clear understanding as to what the Consul really wanted. His Highness had already given him the skin and bones of hippopotami, and many other animals alive and dead. If he wished for any birds, he was welcome to as many as he pleased!

It so chanced, however, that Abbas Pasha took it into his head, or somebody told him, that we had in England several extraordinary breeds of dogs, horses, and cows,—hounds that could catch a gazelle by sheer fleetness, small fighting-dogs that would master a bull,—horses that could compete with his finest Arabian steeds, and beat them in a hard day's hunt over rough ground. He bethought himself, therefore, of the hippopotamus. One good turn of this kind might deserve another of a different kind.

“So, Consul,” said the Pasha abruptly one day, when Mr. Murray was dining with him, “so, you want a hippopotamus?”

“Very much, your Highness.”

“And you think that such an animal would be an acceptable present to your Queen and country?”

“He would be accounted a great rarity,” said the Consul; “our naturalists would receive him with open arms—figuratively speaking,—and the public would crowd to pay their respects to him.”

Abbas Pasha laughed at this pleasantry of the Consul. “Well,” said he, “we will inquire about this matter.” He half-turned his head over one shoulder to his attendants: “Send here the Governor of Nubia!” The attendants thus ordered made their salam, and retired.

Anybody, not previously aware of the easy habits of a despotic sovereign, would naturally conclude that the Governor of Nubia was, at this time, in Cairo, and at no great distance from the royal abode. But it was not so. The Governor of Nubia was simply there—at home—smoking his pipe in Nubia. This brief and unadorned order, therefore, involved a post-haste messenger on a dromedary across the Desert, with a boat up the Nile, and then more dromedaries, and then another boat, and again a dromedary, till the Pasha's mandate was delivered. We next behold the Governor of Nubia, in full official trim, proceeding post-haste with his suite across the Desert, and down the Nile, travelling day and night, until finally he is announced to the Pasha, and admitted to his most serene and fumigatious presence. The Governor makes his grand salam.

“Governor,” says the Pasha—and we have this unique dialogue on the best authority—“Governor, have you hippopotami in your country?”

“We have, your Highness.”

Abbas Pasha reflected a moment; then said—“Send to me the Commander of the Nubian army. Now, go!”

This was the whole dialogue. The Governor made his salam, and retired. With the same haste and ceremony, so far as the two things can be combined, he returned to Nubia by boat, and dromedary, and horse, and covered litter; and the same hour found the Commander of the army of Nubia galloping across the Desert with his attendants, in obedience to the royal mandate.

The Pasha, knowing that all means of speed will be used, and what those means will be, together with the nature of the route, is able to calculate to a day when the Commander ought to arrive—and therefore *must* arrive,—at his peril, otherwise. The British Consul is invited to dine with his Highness on this day.

Duly, as expected, the Commander of the Nubian army arrives, and is announced, just as the repast is concluded. He is forthwith ushered into the presence of the sublime beard and turban. Coffee and pipes are being served. The Commander makes his grand salam, shutting his eyes before the royal pipe.

“Commander,” says the Pasha, without taking his pipe from his mouth, “I hear that you have hippopotami in your country.”

“It is true, your Highness; but——”

“Bring me a live hippopotamus—a young one. Now, go!”

This was actually the dialogue which took place on the occasion—and the whole of it. The Commander of the Nubian forces made his grand salam—retired—and returned as he came,—“big” with the importance of his errand,—but also not without considerable anxiety for its result.

Arriving at Dongola, the Commander summoned his chief officers and captains of the Nubian hosts to a council of war on the subject of the hippopotamus hunt, on the result of which—he intimated—several heads were at stake, besides his own. A similar communication was speedily forwarded to the chief officers of the right wing of the army, quartered in their tents at Sennaar. The picked men of all the forces having been

selected, the two parties met in boats at an appointed village on the banks of the Nile, and there concerted their measures for the expedition.

The Commander divided the chosen body into several parties, and away they sped up the Nile. They followed the course of the river, beyond the point where it branches off into the Blue Nile, and the White Nile. Good fortune at length befel one of the parties; but this cost much time, and many unsuccessful efforts—now pursuing a huge savage river-horse, with rifle-balls and flying darts; now pursued by him in turn with foaming jaws and gnashing tusks—all of which may readily be conjectured, from the fact that they did not fall in with their prize till they had reached a distance, up the White Nile, of one thousand five hundred miles above Cairo. In the doublings and re-doublings of attack and retreat, of pursuit and flight, and renewed assault, they must of course have traversed in all, at least two thousand miles.

Something pathetic attaches to the death of the mother of “our hero,”—something which touches our common nature, but which such hunters as Mr. Gordon Cumming would not be at all able to understand. A large female hippopotamus being wounded, was in full flight up the river; but presently a ball or two reached a mortal part, and then the maternal instinct made the animal pause. She fled no more, but turned aside, and made towards a heap of brushwood and water-bushes that grew on the banks of the river, in order (as the event showed) to die beside her young one. She was unable to proceed so far, and sank dying beneath the water. The action, however, had been so evidently caused by some strong impulse and attraction in that direction, that the party instantly proceeded to the clump of water-bushes. Nobody moved—not a green flag stirred; not a sprig trembled; but directly they entered, out burst a burly young hippopotamus-calf, and plunged head foremost down the river banks. He had all but escaped, when amidst the excitement and confusion of the picked men, one of them who had “more character” than the rest, made a blow at the slippery prize with his boat-hook, and literally brought him up by burying the hook in his fat black flank. Two other hunters—next to him in presence of mind and energy—threw their arms round the great barrel-bellied infant, and hoisted him into the boat, which nearly capsized with the weight and struggle.

In this one circumstance of a hippopotamus being ordered by his Highness Abbas Pasha, has been pleasantly shown the ease and brevity with which matters are managed by a despotic government. We complain at home—and with how much reason, everybody knows too well—of the injurious and provoking slowness of all good legislative acts; but here we have a beautiful little instance, or series of little instances, of going rather too fast. Things are settled off-hand in the East by a royal mandate—from the strangling of a whole seraglio, to the suckling of a young hippopotamus.

Returning down the Nile with their unwieldy prize, for whose wounded flank the best surgical attendance the country afforded, was of course procured, it soon became a matter of immense importance and profound consultation as to how and on what the innocent young monster should be fed. He would not touch flesh of any kind; he did not seem to relish fruit; and he evidently did not, at present, understand grass. A live fish was put into his mouth, but he instantly gave a great gape and allowed it to flap its way out again and fall into the water. Before long, however, the party reached a village. The Commander of the army saw what to do. He ordered his men to seize all the cows in the village, and milk them. This was found very acceptable to their interesting charge, who presently despatched a quantity that alarmed them, lest they should be unable to keep up the due balance of supply and demand. The surplus milk, however, they carried away in gourds and earthen vessels. But they found it would not keep: it became sour butter, and melted into oil. They were, therefore, compelled, after a milking, to carry off with them one of the best cows. In this way they returned fifteen hundred miles down the Nile, stopping at every village on their way—seizing all the cows and milking them dry. By these means they managed to supply the “table” of the illustrious captive, whose capacities in disposing of the beverage appeared to increase daily.

The hunting-division of the army, headed by the Commander-in-Chief, arrived at Cairo with their prize on the 14th of November, 1849. The journey down the Nile, from the place where he was captured, *viz.*, the White Nile, had occupied between five and six months. This, therefore, with a few additional days, may be regarded as the age of our hippopotamus on reaching Cairo. The colour of his skin, at that time, was for the most part of a dull, reddish tone, very like that (to compare great things with small) of a naked new-born mouse. The Commander hastened to the palace to report his arrival with the prize to his royal master, into the charge of whose

officers he most gladly resigned it. His Highness, having been informed of the little affair of the succession of “cows,” determined to place the vivacious un-weaned “infant prodigy” in the hands of the British Consul without a moment’s delay.

The announcement was accordingly made with oriental formality by the chief officer of Abbas Pasha’s palace, to whom the Honorable Mr. Murray made a suitable present in return for the good tidings. A lieutenant of the Nubian army, with a party of soldiers, arrived shortly after, bringing with them the animal, whose renown had already filled the whole city. He excited full as much curiosity in Cairo, as he has since done here, being quite as great a rarity. This will be easily intelligible when the difficulties of the capture, and the immense distance of the journey are taken into consideration, with all the contingencies of men, boats, provisions, cows, and other necessary expenses.

The overjoyed Consul had already made all his preparations for receiving the illustrious stranger. He had, in the first place, secured the services of Hamet Safi Cannana, well known for his experience and skill in the care and management of animals. A commodious apartment had then been fitted up in the court-yard of the Consul’s house, with one door leading out to a bath. As the winter would have to be passed in Cairo, proper means were employed for making this a warm, or tepid bath. Here then our hippopotamus lived, “the observed of all observers,” drinking so many gallons of milk a day (never less than twenty or thirty quarts) that he soon produced a scarcity of that article in Cairo. Nor will this be so much a matter of surprise, when it is considered that they do not understand there the excellent methods of manufacturing enough milk to answer any demand, which obtains with us in London, where such an event as a scarcity of milk was never known by the oldest inhabitant.

Meanwhile active preparations were making for his arrival in Alexandria, to be shipped on board the Ripon steamer. The vessel was furnished with a house on the main-deck, opening by steps down into a tank in the hold, containing four hundred gallons of water. It had been built and fitted up at Southampton from a plan furnished by Mr. Mitchell, Secretary of the Zoological Gardens in the Regent’s Park, to whose energies and foresight we are indebted for the safe possession of this grotesque, good-tempered and unique monster. The tank, by various arrangements, they contrived to

fill with *fresh* water every other day. A large quantity was taken on board in casks; a fresh supply at Malta; and, besides this, which was by no means enough, they made use of the condensed water of the engines, which amounted to upwards of three hundred gallons per day. As there are some hippopotami who enjoy the sea on certain coasts of the world, it is not improbable but our friend would soon have got used to sea-water; but Mr. Mitchell was determined to run no risks, prudently considering that, in the first place, the strength of the salt water, to one whose mother had been accustomed, and her ancestors for generations, to the mild streams of Nilus, might disagree with “young pickle;” and secondly, if he chanced to take to it amazingly, how would he bear the change when he arrived at his mansion in the Regent’s Park. Fresh water, therefore, was provided for his bath every other day throughout the voyage.

The British Consul began to prepare for the departure of his noble guest at the end of April; and in the early part of May, the Consul took an affectionate leave of him, and would have embraced him, but that the extraordinary girth of his body rendered such a demonstration impossible.

So, our hippopotamus departed from Grand Cairo in a large padded cart. He had refused a very nice horse-box which the Consul had provided for him. Some feeling about his dignity, we suppose; though Hamet Safi Cannana considered the objection arose from a certain care of his skin, which might have got a little chafe or hard rub in the horse-box. It was a lesson to Mr. Murray for life. No effort, of course, was made to compel the great personage to enter this machine, because it is one of Hamet’s principles of management never to irritate an animal—always to keep him in good temper—never, directly and immediately to thwart his will in anything that is not injurious, impracticable, or particularly unreasonable. Very delightful all this! Who would not be a hippopotamus? Who that was not Caesar, would not wish to be Pompey?

On arriving at Alexandria, full ten thousand people rushed out into the streets to see our hippopotamus pass. If no one had ever seen the amphibious prodigy in Cairo, it is not to be wondered at that the mental condition of Alexandria was in the same lamentable degree of darkness.

The crowd was so great, that the British Consul (whose feelings had so mastered him on taking leave of his guest, that he had been obliged to follow the *cortége*) was under the necessity of applying to the Governor of

Alexandria for an escort of troops. This was forthwith granted, and down they came galloping along the streets of Alexandria, with waving scimitars! It was well the hippopotamus did not see them from his padded cart, where he lay asleep—it might have caused a little misunderstanding.

Order being restored, and a great lane made in the crowd, Hamet Safi Cannana commenced the gradual and delicate process of awaking the great personage. In the course of an hour or so, during which time the escort of soldiers all “stood attention,” the excited feelings of the anxious lane of population were gratified by the sight of the Arab ceremoniously advancing in gentleman-usher fashion, while close behind him slowly lounged the hippopotamus.

He embarked on board the Ripon, where he was soon joined by his Excellency General Jung Bahadoor Ranajee, and the Nepaulese princes, his brothers. These latter personages would have been great objects of attraction under any other circumstances; but what could stand against such a rival as the occupant of the great house and bath on the main-deck?

During the voyage, “our fat friend” attached himself yet more strongly to his attendant and interpreter, Hamet; indeed, the devotion to his person which this assiduous and thoughtful person had manifested from his first promotion to the office, had been of a kind to secure such a result from any one at all accessible to kindly affections. Hamet had commenced by sleeping side-by-side with his charge in the house at Cairo, and adopted the same arrangement for the night during the first week of the voyage to England. Finding, however, as the weather grew warmer, and the hippopotamus bigger and bigger, that this was attended with some inconvenience, Hamet had a hammock slung from the beams immediately over the place where he used to sleep—in fact, just over his side of the bed—by which means he was raised two or three feet above his usual position. Into this hammock got Hamet, and having assured the hippopotamus, both by his voice, and by extending one arm over the side so as to touch him, that he was there as usual at his side, and “all was right,” he presently fell asleep. How long he slept Hamet does not know, but he was awoke by the sensation of a jerk and a hoist, and found himself lying on the bed in his old place, close beside our fat friend. Hamet tried the experiment once more: but the same thing again occurred. No sooner was he asleep than the hippopotamus got up—raised his broad nose beneath the heaviest part of

the hammock that swung lowest, and by an easy and adroit toss, pitched Hamet clean out. After this, Hamet, acting on his rule of never thwarting his charge in anything reasonable, abandoned the attempt of a separate bed, and took up his nightly quarters by his side as before.

As for the voyage, it was passed pleasantly enough by the most important of the illustrious strangers on board. His Excellency the Nepaulese ambassador, together with the prince his brother, were uncommonly seasick; but as for our fat friend, he enjoyed himself all the way. He liked his bath, for which there was no lack of fresh water supplies, and his provisions were equally satisfactory. Two cows and ten goats had been taken on board for his sole use and service; these, however, not being found sufficient for a “growing youth,” the ship’s cow was confiscated for the use of his table; and this addition, together with we forget how many dozen sacks of Indian corn meal, enabled him to reach our shores in excellent health and spirits.

A word as to the title of “river-horse,” when taken in conjunction with his personal appearance, his habits, and his diet. The hippopotamus has nothing in common with the horse; he seems to us rather an aquatic pig, or a four-footed land porpoise. In fact, he appears to partake of the wild boar, the bull, and the porpoise—the latter predominating at present, but when he gets his tusks, we much fear there will be an alteration in his manners for the worse. As to his eventual size, the prospect is alarming. He is at present only seven months old, and he will continue growing till he is fifteen years of age. What news for the London cows!

Arrived at Southampton, our hippopotamus, house and all, with Hamet Safi Cannana at his side, was hoisted up at the vessel’s yard-arm, and gradually lowered upon a great iron truck, which was then wheeled off to the railway station. The whole concern was deposited in the special carriage of a special train, and on this he travelled from Southampton to London. He arrived at the Zoological Gardens in the Regent’s Park at ten o’clock at night, and found Lord Brougham, Professor Owen, Thomas Bell, and Mr. Mitchell all waiting (we believe they were not in court dresses) to receive him. They were presently joined by the learned Editor of the “Annals of Natural History,” the learned Editor of the “Zoologist,” in company with Mr. Van Voorst, and several artists who made sketches by the light of a lanthorn. Doyle, Wolff, Harrison Weir, Foster, (for the “Illustrated London

News”) and others, were all in assiduous attendance, watchful of every varying outline. The illustrious stranger descended from his carriage, and entered the gardens. First went the lanthorn; then Hamet Safi Cannana with a bag of dates slung over his shoulder; and after him slowly lounged our uncouth treasure, with a prodigy of a grin such as he alone can give, expressive of his humorous sense of all the honours and luxuries that awaited him.

We understand it is a cabinet secret, that the Pasha has ordered a fresh party of hunting soldiers to proceed up the river, as far as the White Nile, to search for another young hippopotamus—a female! We may, therefore, look forward to the unrivalled fame of possessing a royal pair—“sure *such* a pair” as were never yet seen in any collection of Natural History—to say nothing of the chance of a progeny. These are national questions,—why should they be cabinet secrets?

We are certainly a strange people—we English. Our indefatigable energies and matchless wealth often exhibit themselves in eccentric fancies. No wonder, foreigners—philosophers and all—are so much puzzled what to make of us. They point to the unaided efforts of a Waghorn, and to his widow’s pension-mite—and then they point to our hippopotamus! Truly, it is not easy to reply to the inference, and impossible to evade it. We have had a Chaucer and a Milton, a Hobbes, and a Newton, a Watt and a Winsor; and we have had other great poets, and philosophers, and machinists, and men of learning and science, and have several of each now living among us: but any amount of a people’s anxious interest, which the present state of popular education induces, is very limited indeed compared to that which is felt by all classes for a Tom Thumb, a Jim Crow, or our present Idol. Howbeit, as the last is really a great improvement on the two former fascinating exotics, it is to be hoped that we shall, in course of time, more habitually display some kind of discrimination in the objects of our devotion.

CHIPS.

RAILWAY COMFORT

In all the utilities of Railway travelling, England is supreme. Speed, represented by from thirty to sixty miles an hour, “just (to quote the words of Lubin Log) as the passenger pleases;” punctuality, that admits of the setting of watches by arrivals and departures; and safety, exemplified by the loss of no human life from any other cause than the carelessness of the sufferer, during the past two years, are proofs of British supremacy in locomotion. Yet—by a strange perversity not easily accounted for in a country known all over the rest of the world as the Kingdom of Comfort—the point apparently aimed at is to render the transit of the human frame as uncomfortable an operation as possible. Every elegance and luxury is bestowed upon waiting-rooms where extreme punctuality renders it unnecessary for people to wait; and upon refreshment-rooms in which travellers are allowed ten minutes to scald themselves with boiling coffee, or to choke themselves with impossible pork-pies; but carriages in which travellers have to be cramped up, often for hours, and sometimes for whole days, are apparently contrived to inflict as much torture as practicable. In order to force those who cannot afford it into the first-class, second and third-class carriages are only one and two degrees removed from cattle pens. And that these should not be too delicious, the humbler order of passengers will not easily forget that a director once proposed to hire a number of chimney-sweeps to render—what, with the best company, are nothing better than locomotive hutches—perfectly untenable.

They manage these things better abroad. There a detestable class-feeling—a contemptible purse-worship, which rigidly separates people according to their pecuniary circumstances; which metes out the smallest privilege or

comfort at a price—does not exist to prevent the managers of railways from making the journeys of their customers and supporters as pleasant as possible. On the French railroads, (setting aside the question that the fares are much lower,) the second-class carriages are comfortably cushioned, having pretty silk blinds to keep out the sun; windows that really are capable of being pulled up and down, besides hooks for hats,—a great convenience on a journey. For the blinds, indeed, an enterprising blind-maker in France agreed to furnish them to one railway company, gratis, on condition that they used no other for a certain number of years, and allowed him to make them the medium of his advertisements. Talk of advertising vans—can they be compared to the brilliant notion of advertising railways—trains of puffs, wafting the genius of inventors faster than the wind! We throw out the hint to the “advertising world” in this country.

In winter, even in an English first-class carriage, there is no protection against frost and damp; but in nearly all the foreign railways, no sooner does the winter set in than the first-class traveller finds the bottom of his carriage provided with a long tin case full of hot water. In the cold months, masses of woollen cloth and railway wrappers, are seen shaking in the corners of first-class English carriages with shivering, comfortless, human beings inside them, despairing of any sort of warmth whatever.

Comfort in railway travelling is, however brought to the highest perfection in Germany. An esteemed correspondent at Vienna writes to us on this subject in the following terms:—On the “*Wiener-Neustädter Eisenbahn*,” (the Vienna and Neustadt Railway), the carriages of the first, second, and third-class may each be said to resemble a spacious room, furnished with seats, something like a concert-room, and having a broad passage down the middle. Thus one may get up, walk towards a friend a dozen seats off; or, if you require more air or a change of position, you will find the backs of the seats shift so as to enable you to turn round, and sit down the other way without inconvenience to any one. I need not say that on this railway there is no struggle for “that corner place with your back to the engine,” which is a desirable object throughout our three kingdoms,—for every place is a corner place, having light and air, and you may sit which way you please.

Attached to each carriage, and going the whole length of the train, is a broad wooden plank, along which the guards are constantly walking, so that

the slightest thing amiss could scarcely occur without their perceiving it immediately. Just before the arrival of the train at any station, one of these functionaries—for there are several—quietly opens the door and, instead of calling out “I say, you sir!” or “Come, marm, your ticket, I can’t be a waitin’ here all day,” as we have heard in England, walks without any hurry or bustle down the division from one end to the other, repeating, in a clear and ordinary tone of voice the name of the station which is being approached, and requiring the tickets of such passengers as are going to alight there. With such an arrangement—giving ample time for the gathering together of coats, canes, umbrellas, reticules, and so forth—even Martha Struggles herself might have got through a journey unscathed and “unflustered.”

The admirable arrangement displayed in America, as well as in Germany, for receiving tickets without that delay which has been so much complained of in England, cannot be sufficiently applauded. When however delay is unavoidable, to receive the mails, or from some other cause, no sooner does the train stop, than a waiter, or sometimes a pretty waitress—who is more likely to find customers—trips up the steps with a tray laden with iced water and lemonade, glasses of light wine or *maitrank*, (a kind of Burridge-cup,) biscuits, cakes, and other edible nick-nacks, so that the passenger may take some slight refectation without getting down.

In the railway from Bonn to Cologne, on the Rhine, they have pushed convenience yet farther, having provided the first-class carriages with tables, so that during the journey, one pressed for time may write letters with the greatest ease; pens and a portable inkstand being all that is necessary for that purpose. Paper may be had at the station.

It has been also suggested on several of the continental railways, that such travellers as chose to pay for the space, might have a regular bed; a great convenience for ladies or invalids, unable to bear the fatigue of a journey of many hours by night.

These hints might be followed with very great advantage to the shareholders in particular and to the public in general, by the directors of British lines.

IMPROVING A BULL.

The highly respectable old lady who addressed us on a former occasion, has obliged us with another communication, on a most important subject:—

“Sir,—You would have heard before, but the cause was a mad bull, which being tossed might at my age be very ill-convenient. But that’s nothing to what I’m going to tell you. Only to think of the power of horns! Bulls tosses very high, I’ve heard, but did you ever hear, Mr. Conductor, of a mad bull tossing a widow and six children across the sea, half over the side of the round world, from our Borough to Australia? Well you may stare, but it’s a fact!

“The bull run right at me, full butt, and so I grasped my umbrella with both hands and ran to where the shops was—drat the boys, how they did screech about one!—and it was cold water, which I doesn’t often drink, by which means I came to in a pastry-cook’s. The name was Bezzle, I see it on a bag while she was putting in gingerbread nuts for Mrs. Jenks’s baby, which I bought not to be under obligation for stepping in.

“‘Gracious mussy, Mrs. Bezzle,’ says I, ‘why wasn’t I killed? What ever is the reason of them bulls?’

“Says she, ‘It’s market day.’

“‘Smithfield!’ says I.

“Says Mrs. Bezzle, ‘Mum, all the abuse and outcry against Smithfield is very narrow-minded.’

“Says I, ‘How so?’

“Says she, ‘It don’t consider shop-keepers. When a bull takes a line of street, it drives the people into the shops on either side, and they make purchases for fear of being gored.’

“‘Heighty teighty, mum,’ I says, ‘you are alluding to my gingerbread.’

“Says she, ‘I scorn allusions. It’s a rule. Whether it’s bulls or thunderstorms, or what it is we look to, we respects whatever sends us customers.’

“Says I, ‘Mrs. Bezzle, you astonish me. Where’s your family trade?’

“Says she, ‘There are too many traders. Where one of us earns meat, three of us only earn potatoes.’

“‘Emigrate,’ says I.

“Says she, ‘That’s very well, but then,’ says she, ‘in such a move it’s hard to know which way to put one’s foot, and when a step’s made, if it’s a wrong one, it’s not easy to retrace it.’

“‘Spirited trading—’ says I.

“‘Ah!’ says she, cutting me short rudely; but I forgive her, owing to her feelings. ‘Take Chandlery, within seven minutes of this door, mum. One man sells soap under cost price, and other things at profit, hoping to bring people to his shop for soap, and then get them to buy other articles. But his neighbour sells cheap herrings in the same way; another sacrifices pickles, and another makes light of the candle business. What’s the result? Folks buy in the cheapest market; go for soap to the man who sells that at the ruin prices, go for herrings to his neighbour, go down the other street for pickles, and get candles over the way.’

“‘Well,’ says I, ‘that’s an Illustration of Cheapness, but,’ says I, ‘it’s dishonest. A fair trader has no right to sell an article at less than its first cost.’

“‘No right!’ says she. ‘And I dessay he thinks he has no right to starve. It’s very hard to judge. The young tradesman, with his little capital and knowledge of a trade, has got his sweetheart and his ambition. He must wedge into society somehow, and he begins with the sharp end.’

“‘But,’ says I, ‘it isn’t sharp, Mrs. Bezzle.’

“So she shakes her head; says she, ‘I’ll give you an example which is true, and one out of a many.’

“‘I once knew an excellent young man who died of cholera. He left a widow and three little children. After deducting all expenses for her husband’s burial, the widow found that she possessed a hundred pounds. With fear and trembling, she embarked this money, in an effort to support

herself. With it she fitted up a little shop, and had begun to earn a livelihood, when——’

“‘Well, Mrs. Bezzle, what prevented her?’

——“‘An empty house close by was taken by another person following her trade. Immediately her receipts diminished. One cannot live except by bread that can be got out of a neighbour’s cupboard. The widow and the children have already lost eighty pounds, have only twenty left; their house is taken by the year, and so they still are in it; and the poor lost woman cannot be comforted. Her hope is gone.’

“‘Heigh, dear,’ says I, ‘it wasn’t so in my young days. I believe this is owing to overpopulation,’ says I.

“‘Well,’ says Mrs. Bezzle, perking up. ‘It’s cruel to blame us for our struggles. What if I *have* got nine, and six on ’em dependant on penny tarts and gingerbread for meat, drink, washing, and lodging, are they to be thrown in my teeth?’

“‘Emigrate,’ says I, six times more pointedly than before.

“‘Where to?’ says she, ‘and how? Who can tell me that?’

“‘Go and lay your case before Parson Pullaway; he knows our M.P., and *he* knows all about colonial places. Hasn’t his brother’s wife’s first cousin got one of them? He is Sub-under-Secretary to Lord Oxfordmixture, who has all the emigration settlements under his thumb.’

“‘I’ll think about it,’ says Mrs. Bezzle, quite struck-like,—for down came the scales on the counter like a shot, and the whole ounce of sugar-candy jumped into the little boy’s apron of its own accord. He had come for two penn’orth on pretence of a cough. ‘Besides, didn’t Mr. Pullaway christen seven out of my nine children, and not a penny of the fees owing for?’

“‘The last word as ever I spoke to Mrs. Bezzle was, ‘Emigrate!’

“‘Well, who would have thought it? Next week Mrs. Bezzle’s business was to sell. The week after, it was sold. The week after that, Mrs. Bezzle and her son Tom, and Tom’s wife, and Tom’s brother Sam, and Mrs. Bezzle’s eldest daughter, and little James, and Sarah, and Mary Ann, and the two little urchins, were on board a ship, at Liverpool, bound for Port Philip. That’s a year, come Michaelmas, ago.

“‘But, drat ’em, why didn’t they pay the postage? Two-and-two is a consideration when butter (best fresh) is a rising a penny a pound every

week. Not but what I was glad to hear from Mrs. Bezzle. Tom and his wife, and his brother Sam, are settled in a 'run;' and though there was some words I couldn't make out, I dare say they didn't explain how a 'run' could be a settlement. 'Quite the reverse!' as Mrs. Jenks said—(I have made it up with her, though she did insinuate the gingerbread nuts the mad bull made me buy gave her babby the cholera; and, bless it! it was only the teeth after all). Mrs. Bezzle has settled herself in the mutton-pie and cheesecake line, and has no fear of opposition; and as in Port Philip there is good digestions and plenty of 'em, pies is popular. Prices, too, is better,—penny pies being tuppence. James is on the 'run,' along with his eldest brother. Sarah an't married yet,—for out of six offers, a young gal of seventeen has a right to be puzzled for six months or so, and more dropping in every week. Mary Anne is family governess to a rich copper-man, with plenty of stock—I suppose by that he is in the soup line. However, all is doing well.

“Well, Mr. Conductor, it was all owing to that bull, wasn't it? If I hadn't improved that solemn occasion, where would Mrs. Bezzle, and four out of six of her helpless offspring have been by this time?—why, in the workhus.”

LUNGS FOR LONDON.

Travellers describe nothing to be so much dreaded by the people of the East as a flight of locusts, except indeed a settlement of locusts. When those devouring insects alight on the fields and pastures, they begin from a centre composed of myriads, and eat up everything green within radii extending over not acres, but miles. They fall upon gardens and leave them deserts; and upon a field they do not permit so much as a blade of grass to indicate where grass was.

Although, in fact, these little devastators do not trouble us; in effect, Londoners are the victims of equally efficient destroyers of their green places.

Bricklayers are spreading the webs and meshes of houses with such fearful rapidity in every direction, that the people are being gradually confined within narrow prisons, only open at the top for the admission of what would be air if it were not smoke. Suburban open spaces are being entombed in brick-and-mortar mausoleums for the suffocation as well as for the accommodation of an increasing populace; who, if they wish to get breath, can find nowhere to draw it from, short of a long journey. The Lungs of London have undergone congestion, and even their cells are underground.

Of all the neighbourhoods of which London is a collection, Finsbury and Islington have suffered most. Within the recollection of middle-aged memories, Clerkenwell Green was of the right colour; Moorfields, Spafields, and the East India Company's Fields, were adorned with grass; and he must be young indeed who cannot remember cricket-playing in White Conduit, Canonbury, Shepherd and Shepherdess, Rhodes, and Laycock's, besides countless acres of other "Fields," which are now blotted out from the face of the Country to become Town, in the densest sense of

the word. Thanks to the window tax and the bricklayer, fresh air will be thoroughly bricked out, unless a vigorous effort be made to stop the invasion of burnt clay and water.

Mr. Lloyd, a gentleman of Islington who dreamt a few years since that he lived in the country, but has recently awoke to the conviction that his once suburban residence has been completely incorporated with the town, determined, if possible, to arrest the invasion of habitations. His plan is to dam out the flood of encroachment by emparking a large space at Islington for the behoof of the Borough of Finsbury, which contains a population of three hundred thousand panting souls. This space is, according to his plan, that which surrounds the village of Highbury, one of the highest and airiest suburbs of London. It is within two miles of the City, and might be rendered accessible to Victoria Park in the east, and to Regent's Park in the west. The proposed enclosure will take in a good portion of the course of the New River, and a large quantity of ground so well and picturesquely wooded, that a paling and a name are only requisite to convert it at once into a park. In shape the enclosure would be a triangle, the base of which is the Holloway Road and Hopping Lane, and the apex, a point at which the Seven Sisters' Road joins the Green Lanes. The extent of these grounds is about three hundred acres, and the total cost of securing them to the public is not more than one hundred and fifty thousand pounds.

Mr. Lloyd has been vigorously agitating this matter for more than nine years, and yet—such is the pace at which the public are apt to move in affairs in which the public alone is itself concerned—it is only lately that he has obtained an attentive hearing for his plan.

A prospect of success appears now, however, to dawn. Public meetings have been lately held in every district concerned, in which every sort of co-operation has been promised. A single difficulty seems to stand in the way; one little thing needful is only required to turn the project into an accomplished fact, and that is, the money,—one hundred and fifty thousand pounds merely. Mr. Lloyd and his coadjutors have, we believe, mentioned their little difficulty at the Treasury, and are awaiting an answer. This state of things would form a curious problem for De Morgan, Quetelet, or others learned in the doctrine of probabilities: given, official routine multiplied by systematic delay, what are the chances of the cash required within the present generation?

A park for Finsbury is too urgent a demand for a dense population to allow of much time being wasted in knocking at the door of the Treasury. The public must bestir *themselves* in the scheme, and it will soon be accomplished and carried out.

THE LOVE OF NATURE.

Where the green banners of the forest float,
Where, from the Sun's imperial domain,
Armour'd in gold, attentive to the note
Of piping birds, the sturdy trees remain,
Those never-angered armies; where the plain
Boasts to the day its bosom ornaments
Of corn and fruitage; where the low refrain
Of seaside music song on song invents,
Laden with placid thought, whereto the heart assents,
Often I wander. Nor does the light Noon,
Garrulous to man's eye, declaring all
That Morning pale (watched by her spectre moon,
Or solemn Vesper, seated near the pall
Of Day) holds unrevealed; nor does the fall
Of curtain on our human pantomime,
The sweeping by of Day's black funeral
Through Night's awe-stricken realms, with tread sublime,
Chiefly delight my heart; beauty pervades all time.
Morning: the Day is innocent, and weeps;
Noon: she is wedded and enjoys the Earth;
Evening: wearied of the world she sleeps.
Night watches till another Day has birth.
The innocence of Morning, and the mirth
Of Noon, the holy calm of Eventide,
The watching while Day is not, there is dearth
Of joy within his soul who hath not cried:
"I welcome all, O God,—share all Thou wilt provide!"

THE PRESERVATION OF LIFE FROM SHIPWRECK.

It is a difficult matter to reconcile with the sympathy, which it is well-known the sufferings of the unfortunate always receive in England, the apparent apathy which exists among the public, on a subject so important as the preservation of Life from Shipwreck. Several pleas in extenuation have been urged by those most interested. In the first place, there is that natural hardihood and contempt of danger in the English sailor, which it is, occasionally, impossible to tame down to anything like prudence and forethought. This indomitable spirit of emulation and daring, is found to be the greatest enemy to the adoption of any of those appliances which science has rendered available. The Deal boatman trusts his life in precisely the same sort of craft that his father, and his father's father, did before him. Confident in, and proud of, the skill which he has inherited from them, he scorns to tarnish, as he falsely reasons, his name by the habitual use of buoy or belt, lest those of his comrades who are firmly entrenched behind their ancient prejudices, should set him down as faint-hearted, and unworthy the honourable name of a "Deal boatman."

The still more inaccessible Scotch fisherman, with his four thousand piscatory brethren, "shoots his nets" on the exposed coast of Caithness, in the open boat used by his ancestors, notwithstanding the evil consequences which have often ensued. The latest example of the ill effects of this tenacity of opinion occurred two years since, when a fearful gale, which did more or less damage along the whole eastern face of England and Scotland, wrecked and damaged a hundred and twenty-four of their boats, drowned a hundred men, and occasioned a loss to the fishing community of above seven thousand pounds, which, although a large sum, will not bear any

comparison with the misery and destitution thus entailed upon the widows and orphans of the lost.

It is impossible to say how many of these unfortunate men might have been saved, had they had proper harbours to run for, with lights and beacons to warn, and life-boats to afford assistance; proper boats to keep the sea, and buoys and belts, as a last resource; but surely we are warranted in thinking that fully one half would have been left among us.

In both these examples, it must be acknowledged that it would be a useless effort to attempt any sudden innovations on these deeply-seated prejudices; the only thing that can be done, in either case, is to let the new principle quietly work of itself. Let us find a life-belt for the Deal boatman, which he can wear and work in, until in it he recognises his best friend; let the Scotch fisherman have ocular demonstration that the “model” boat prosecutes the fishery with equal success, and far greater safety and comfort in bad weather, and we shall soon have a different system of things.

In the course of each year an average of something like six hundred ship disasters occur on the shores of this kingdom alone,—some wrecked through stress of weather; some by carelessness, and other disgraceful causes; some through mistaking lights, or having been lured to destruction by useless ones; some through actual rottenness of timber; some dashed to pieces on the very rock for which they were anxiously looking half a mile further a-head, where it *ought* to have been, according to the chart; and some from other causes, more or less easily averted. These losses are attended by the almost incredible destruction of a thousand lives, and the value of tens of thousands of pounds sterling.

The shocking wreck of the Orion—not, we say with sorrow, the last occurrence of the kind—startled, for a moment, the public from their culpable apathy. But the shock passed away; and attention to this subject is gradually subsiding into the usual indifference. The details of this catastrophe ought to have had a more permanent effect on the public mind. In the moment of danger, the gear of the boats was so imperfect, that these could only be released from their davits by capsizing their human cargoes into the deep. Even when they righted, they immediately filled, for the plug-holes were actually unstopped. The most ordinary precautions for saving life were not at hand, as precautions. The hen-coops, barrels, seats,

combings, and other means of escape, by which many were saved, were purely accidental life-preservers.

Every English ship, before leaving port, should be submitted to a supervising power similar to the inspection that emigrant ships undergo, in order that it should be certified that means, both simple and efficacious, for the safety of the passengers and crew, exist on board—boats, belts, mattresses, rafts; everything, in short, that can add to the security of those about to “go down to the sea in ships.”

That this sort of supervision is effectual, is proved by the few disasters which happen to the vessels of the Royal Navy. In these ships, everything is not only kept in its proper place, to be ready when wanted, but each man is constantly exercised in what he is to do with it when no danger is apprehended, that he may be in a state of prompt efficiency when it is. The Commander-in-Chief of the Mediterranean squadron can step on board any one of his ships in the middle of the night; and although three-fourths of its crew are asleep in their hammocks, he can, by ordering the “beat to quarters,” make sure of every man being at his post in seven minutes, ready for action or for any sudden disaster. This sort of discipline it is which is so much required in the merchant navy. In case of a ship striking, a dozen men rush to do one thing,—perhaps to release a boat from one of her davits,—and, consequently, swamp the boat, by leaving the stern rope untouched. Captain Basil Hall, in his “Fragments of Voyages and Travels,” describes the vigilant precaution daily made even against the loss of one life. To each life-buoy there is as regular a “service” as to any other part or apparatus of the ship. He says:—

“On the top of the mast is fixed a port-fire, calculated to burn, I think, twenty minutes or half-an-hour; this is ignited most ingeniously by the same process which lets the buoy down into the water. So that a man falling overboard at night, is directed to the buoy by the blaze on the top of its pole or mast, and the boat sent to rescue him also knows in what direction to pull. Even supposing, however, the man not to have gained the life-buoy, it is clear that, if above the surface at all, he must be somewhere in that neighbourhood; and if he shall have gone down, it is still some satisfaction, by recovering the buoy, to ascertain that the poor wretch is not left to perish by inches. The method by which this excellent invention is attached to the ship, and dropped into the water in a single instant, is perhaps not the least

ingenious part of the contrivance. The buoy is generally fixed amidships over the stern, where it is held securely in its place by being strung, or threaded, as it were, on two strong perpendicular iron rods fixed to the taffrail, and inserted in holes piercing the framework of the buoy. The apparatus is kept in its place by what is called a slip-stopper, a sort of catch-bolt or detent, which can be unlocked at pleasure, by merely pulling a trigger. Upon withdrawing the stopper, the whole machine slips along the rods, and falls at once into the ship's wake. The trigger which unlocks the slip-stopper is furnished with a lanyard, passing through a hole in the stern, and having at its inner end a large knob, marked 'Life-Buoy;' this alone is used in the day-time. Close at hand is another wooden knob, marked 'Lock,' fastened to the end of a line fixed to the trigger of a gun-lock primed with powder: and so arranged, that when the line is pulled, the port-fire is instantly ignited, while, at the same moment, the life-buoy descends, and floats merrily away, blazing like a lighthouse. It would surely be an improvement to have both these operations always performed simultaneously, that is, by one pull of the string. The port-fire would thus be lighted in every case of letting go the buoy; and I suspect the smoke in the day-time would often be as useful in guiding the boat, as the blaze always is at night. The gunner who has charge of the life-buoy lock sees it freshly and carefully primed every evening at quarters, of which he makes a report to the captain. In the morning the priming is taken out, and the lock uncocked. During the night a man is always stationed at this part of the ship, and every half-hour, when the bell strikes, he calls out 'Life-buoy!' to show that he is awake and at his post, exactly in the same manner as the lookout-men abaft, on the beam, and forward, call out 'Starboard quarter!' 'Starboard gangway!' 'Starboard bow!' and so on, completely round the ship, to prove that they are not napping."

We should like to hear of Government experimenting with rockets and mortars, with a view to their improvement. Often the safety of a whole ship's company has depended upon the strength of a light cord, attached to a rocket, which has been lying in store for years; often it has happened that this very cord has been *just* a few feet too short! or has snapped, or has got entangled, or something else equally simple, but equally fatal. Let us look also to our *quasi* life-boats, some so heavy that they cannot be launched, or so dangerous as to drown their own crews—some constructed one way, some another—none on any recognised and universal principle. We are very

proud of our name of Englishmen, and lay the flattering unction to our soul, that we are a highly civilised and reasonable community; but whilst we grow magniloquent in praises of our country and her commerce, we forget that we owe it all to the poor Jack Tar, for whose life and comfort we don't seem to care a fig. Else why have these inquiries not been before instituted? What is the use of our Trinity Boards, and Ballast Boards, and Lighthouse Boards, and all other Boards, if the seaman is not to know one light from another when he sees it, or if it is to be placed so that he *cannot* see it? What is the use of our keeping up a Hydrographic department, at an expense little short of thirty thousand a-year, if the surveys, and charts, and valuable data, the result of its labours, are to be so little appreciated? The truth is, that the masters of many of the mercantile marine are incapable of taking advantage of them, and of other improvements in nautical science, from incompetence. We trust, however, that the bill intended to remedy *that* defect, lately introduced by the Ministry into the House of Commons, will, if passed, have the desired object. Although it has been abandoned "at this late period of the session" out of respect to the approaching 12th of August and 1st of September, we trust it will be taken up again soon after the next meeting of Parliament.

WINGED TELEGRAPHS.

Magnetic Electricity for telegraphic purposes has nearly superseded pigeons. Till very recently a regular “service” of Carrier Pigeons existed between London and Paris, for the quick conveyance of such intelligence as was likely to affect the funds. The French capital was the focus of the system, in exemplification of the adage that “all roads lead to Paris,” and pigeon expresses branched off in all directions from that city even to St. Petersburg. Relays of them are still kept up between Paris and Madrid, besides a few other places. The most celebrated relays of winged messengers were those which bore intelligence between Antwerp, Brussels, and Paris. In the former city a society of pigeon fanciers, for amusement and emulation, keeps up an establishment of them. Their doings are amusingly chronicled in Kohl’s last book of Travels, *Reisen in den Neiderlanden*.

Having been invited to join some members of the Society of Antwerp Pigeon Fanciers, he wended his way about five o’clock one morning through the silent streets of the ancient city. A few members of the association, he says, who directed the expedition, were followed by servants carrying two flat baskets, in which the pigeons, about to be dispatched, were carefully deposited. As we proceeded along, my companions related to me some particulars concerning the carrier pigeons, or “*pigeons voyageurs*,” as these winged messengers are designated. The carriers are a peculiar race of pigeons endowed with powers of memory and observation which enable them to find their way to any place by a course along which they have once flown. Every kind of pigeon is not capable of being taught to do this. Of the methods adopted by the Antwerp association for training and teaching these carriers, I learnt the following particulars.

Supposing a dispatch of pigeons is to be sent off from Antwerp to Brussels or Paris, the birds are kept for some time at the place of arrival or terminus, and during that interval are plentifully fed and carefully tended. By little excursive flights, taken day by day, they are gradually familiarised with different parts of the town in which they have been nurtured, and with places in its vicinity. When sufficiently practised in finding their way to short distances, the pigeons are conveyed to a station some leagues from their dove-cote. Here they are kept for a time without food, and then set to flight. On taking wing, they rapidly soar to a vast height, scanning the line of the horizon to discern the church spires, or other lofty points which enable them to distinguish their home. Some of the less intelligent birds lose their way, and are seen no more. Those who return home (to Paris, or wherever else it may be), are again plentifully fed. Then after a little space of time they are carried in baskets some miles further in the direction of Antwerp; again they are put on a short allowance of food and negligently tended. When the pigeons depart on their next flight, the Parisian church spires have sunk far beneath the horizon; however, they soon succeed in combining that portion of the route with which they are acquainted with the part as yet unknown to them. They hover round and round in the air, seeking to catch one or other thread that is to guide them through the labyrinth. Some find it; others do not.

In this manner the carrier pigeons are practised bit by bit along the whole distance between Paris and Antwerp. They attentively observe, or study, and learn by heart, each conspicuous object which serves them as a land-mark on the way. It is usual to exercise particular pigeons between the two cities, which it is wished to connect by this sort of postal communication; and it is necessary to have a certain number for going, and others for returning. After the birds have been accustomed to inhabit a certain district, and to travel by a particular route, it is not found easy to divert them from their wonted course, and to make them available in any other direction.

My friends, the members of the Antwerp Society, assured me that their pigeons had frequently flown from Paris to Antwerp in six or seven hours; consequently in a much shorter time than that in which the same journey is performed by the railway train. By bird's flight, the distance between the two cities is forty miles (German^[1]), and therefore it follows that these carrier pigeons must travel at the rate of from twenty to thirty English miles an hour. It is scarcely conceivable that they should possess the strength of

wing and vigour of lungs requisite for such a flight; and it is no unfrequent occurrence for several of them to die on arriving at their journey's end. In stormy weather the loss of two-thirds of the birds dispatched on such a long flight, is a disaster always to be counted on. It is, therefore, usual to send off a whole flock, all bearing the same intelligence, so as to ensure the chance of one at least reaching its destination.

[1](#). The German mile includes nearly three and a half English miles.

The pigeon expedition which I saw dispatched from Antwerp, consisted of about thirty birds. The point of departure was a somewhat elevated site in the outskirts of the city. A spot like this is always made choice of, lest the pigeons, on first taking flight, should lose themselves amidst the house-tops and church spires of the city with which they are unacquainted; and by having the open country before them, they are enabled to trace out their own land-marks. When the pigeons are to be sent off on lengthened journeys, it is usual to convey them to the point of departure at a very early hour in the morning:—by this means they are dispatched in quietude, unmolested by an assemblage of curious gazers, and they have the light of a whole day before them for their journey. Carrier pigeons do not pursue their flight after night-fall, being then precluded by the darkness from seeing the surrounding country with sufficient distinctness to enable them to discern their resting-places, or stations. In the obscurity of night the whole flock might light on strange dove-cotes, and be captured; an accident which would occasion the total failure of a postal expedition, for the few pigeons who might escape capture, would, on the return of morning, be bewildered, and unable to recombine their plan of route.

Pigeons are not suited for postal communication between places so remote one from another that the journey cannot be completed in a single day. If it can be accomplished in one flight, so much the better. Antwerp and Paris are, I believe, the extreme points of distance within which carrier pigeons are capable of journeying with certainty.

Herr Kohl gives no account of these stations or stages. We once saw one at Montrieul, the first station beyond Dover, towards Paris. The town stands on a high eminence, and is well adapted for the purpose. The cote was on the roof of a *café*. It was a square apartment with a flat ceiling, in which was cut a small door or trap: on the inside of this was fixed a small bell. If a Dover pigeon had alighted on the trap, the bell would have rung, and called

the attention of an attendant always in waiting. The pigeon would have been secured, the dispatch taken from under its wing, and the messenger put into its cage. In a twinkling the cyphered paper would be fastened under the wing of the Beauvais or Amiens pigeon, and it would be sent off. On arriving at its destination, the same formula would be gone through, and the Paris pigeon would take the dispatch to its destination. Although several pigeons, even in fine weather, are entrusted with the same message, two seldom arrive at the common destination at the same time, so that at each place the operation we have described is frequently repeated, in order that at least one of many dispatches may be certain of arriving at the destination.

These establishments were costly. Besides the great number of pigeons necessary to be kept at each station, some of the single birds were valuable. Fifty and sixty pounds was sometimes given for a clever pigeon. Those between Dover and Montrieul, and *vice versâ*, were among the most valuable, for none but sharp-sighted messengers could find their way across the Channel; few flights were sent away without some members of it being lost.

But to return to the Antwerp pigeons—and to Mr. Kohl. Having, he continues, reached the open, elevated spot before-mentioned, the flat baskets carried by the servants were uncovered, and the little *voyageurs* rapidly winged their way upwards. The intelligence they were to convey to Paris was written in little billets, fastened under their wings. The pigeons I saw sent off had been brought in covered baskets from Paris, and were as yet totally unacquainted with Antwerp and its environs. Their ignorance of the locality was manifest in the wavering uncertainty of their movements when they first took wing. On rising into the air, they gathered closely together, like foreigners in a strange country, and presently they steered their course along the confines of the city, in a direction quite contrary to that of Paris. They then soared upwards, spirally, and after several irregular movements (during which they seemed to be looking for the right way, and hesitating which course to take), they all suddenly darted off to south-west, directing their rapid flight straight to Paris, as if gladly quitting inhospitable Antwerp, where they had been scantily fed and carelessly tended.

As soon as the birds were fairly out of sight, the pigeon-trainers proceeded homeward, not a little gratified by the conviction that their fleet messengers, with the intelligence they bore under their wings, would

outstrip the speed of a railway train which had started some time before them.

To me the most interesting point in the whole scene was the interval (about the space of a quarter of an hour) during which the pigeons wavered to and fro, seeking their way in a state of uncertainty. That appeared to me to be a wonderful manifestation of intelligence on the part of the birds. It is frequently affirmed that the carrier pigeon finds its way without the exercise of intelligence or observation, and merely by the aid of some incomprehensible instinct; but, from my own observations of the Antwerp pigeons, I am convinced that this is a mistake. Another circumstance tending to show that the birds are guided by something more than mere instinct, is, that during foggy weather the employment of carrier pigeons is found to be almost as impracticable as the use of the optical telegraph. But though it is not the practice to dispatch carrier pigeons at times when the atmosphere is very thickly obscured by fog, yet, owing to the keenness and accuracy of the visual power of these birds, which is much more perfect than that of man, they have an advantage over the telegraph. The latter is wholly useless when the atmosphere is only slightly obscured; but carrier pigeons frequently soar quite above the region of mist, and are thus enabled to trace their course without interruption. Stations of carrier pigeons are established in most of the principal towns of Belgium.

The members of the Antwerp pigeon-training society, whom I accompanied on the occasion above described, were citizens of the middle class of society. But in Belgium, pigeon-training has its attractions even for persons of rank and wealth, many of whom are enthusiastic pigeon fanciers; indeed, pigeon-flying is as fashionable an amusement in Belgium as horse racing in England. Prizes, consisting of sums of money as high as sixty thousand francs, are frequently won in matches of pigeons—to say nothing of the betting to which those matches give occasion.

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TRANSCRIBER'S NOTES

- Fixed typos; non-standard spelling and dialect retained.
- Renumbered footnotes.

*** END OF THE PROJECT GUTENBERG EBOOK HOUSEHOLD
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